DSO562 Fraud Analytics

Project 3 Report

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Section 1. Executive Summary

Description of Business Problem

The City of New York faces a significant challenge with property tax fraud, where property owners may intentionally misreport property characteristics to reduce their tax liabilities. This misrepresentation leads to revenue losses for the city and an inequitable tax burden among property owners. To address this issue, the city seeks to develop a sophisticated algorithm capable of analyzing approximately one million property records, each containing 32 fields, to identify anomalies and patterns that deviate from expected norms. By detecting these irregularities, the city aims to target potentially fraudulent properties, ensuring accurate tax assessments, improving fairness in tax distribution, and preserving the city's financial integrity.

Statement of Business Results

The fraud detection analysis produced actionable insights by uncovering transaction patterns and ranking anomalies based on final fraud scores. These scores ranged from near zero to over one million, with a median score around 500,000. Key variables such as transaction frequency, ZIP code-based features, and tax classifications emerged as significant contributors to fraudulent patterns, particularly in the top 10 records with the highest fraud scores. The absence of extreme outliers confirms that the dataset is well-balanced, demonstrating the robustness of the model in reliably detecting potential fraud without being influenced by skewed data. This allows the city to efficiently focus investigations on high-risk properties, reducing false positives and delivering significant cost savings through targeted audits and corrective actions.

Section 2. Description of Data

The dataset comprises approximately 1,070,994 property records from New York City, representing detailed property attributes such as dimensions, assessed values, ownership, and location. These records, sourced from public property valuation and assessment data, aim to provide insights into property characteristics and potential anomalies that may indicate irregularities, such as underreported valuations or tax fraud.

The dataset includes both numeric and categorical fields. Key numeric variables include FULLVAL (the total market value of a property), AVLAND (assessed land value), AVTOT (assessed total value), and property dimensions like LTFRONT (lot frontage), LTDEPTH (lot depth), and STORIES (number of floors). Categorical fields provide further classification, such as TAXCLASS (property tax classification), BLDGCL (building class codes), ZIP (geographic location), and OWNER (ownership details). Each record represents a unique property, identified through fields like RECORD and BBLE.

A few distributions stand out as particularly informative. The FULLVAL variable ranges from $0 to $6.15 billion, with a mean value of approximately $874,265. The distribution shows a bimodal pattern, with a significant number of properties valued near zero and a second peak around $400,000, highlighting variability in market values. Similarly, AVLAND values, which range from $0 to $2.66 billion, are highly skewed, with most properties concentrated below $10,000, suggesting that many records represent small or underdeveloped lots. The STORIES field, which captures the number of floors, reveals that the majority of buildings have fewer than 10 stories, while only a small subset represents high-rise structures exceeding 50 floors.

Geographically, properties are spread across the five boroughs of New York City. Queens and Brooklyn have the highest number of records, while Staten Island and Manhattan contribute fewer properties. This reflects the urban density and land-use patterns across different boroughs. Certain categorical fields, such as TAXCLASS, reveal trends in property taxation. Class 1 properties (primarily residential) dominate the dataset, followed by Classes 2 (cooperatives, condominiums) and 4 (commercial properties).

In summary, this dataset provides a comprehensive snapshot of NYC property records, allowing for detailed analyses of property valuations, geographic distributions, and classification trends. The presence of outliers in fields such as FULLVAL, AVLAND, and STORIES offers a basis for identifying anomalies or irregularities that could warrant further investigation.

Summary Statistics

1. **Numeric Table**

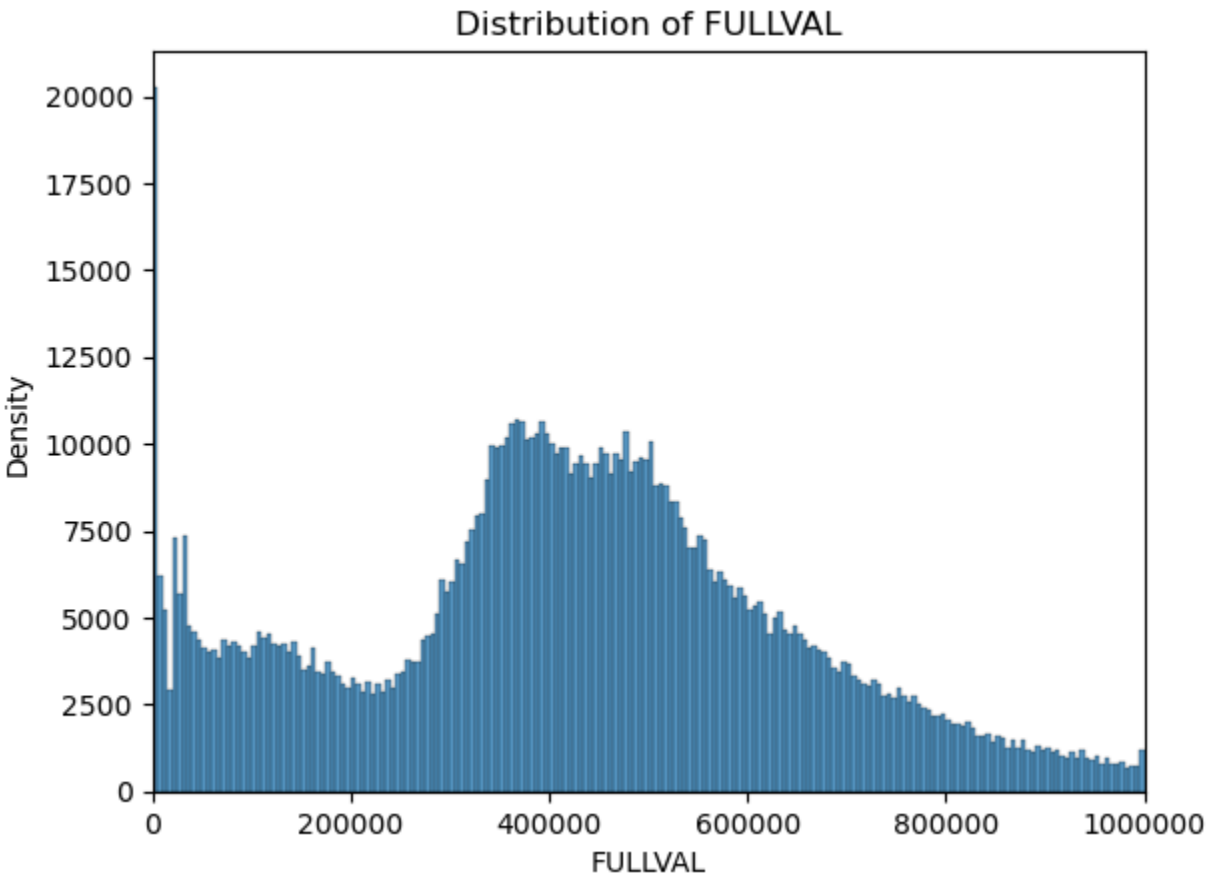
Field Name	# records with values	% Populated	# Zeros	Min	Max	Mean	Std	Most common
LTFRONT	1,070,994	100.0%	169,108	0	9.999e+03	3.664e+01	7.403e+01	0
LTDEPTH	1,070,994	100.0%	170,128	0	9.999e+03	8.886e+01	7.640e+01	100
STORIES	1,014,730	94.7%	0	1	1.190e+02	5.010e+00	8.370e+00	2
FULLVAL	1,070,994	100.0%	13,007	0	6.150e+09	8.743e+05	1.158e+07	0
AVLAND	1,070,994	100.0%	13,009	0	2.668e+09	8.507e+04	4.057e+06	0
AVTOT	1,070,994	100.0%	13,007	0	4.668e+09	2.272e+05	6.878e+06	0
EXLAND	1,070,994	100.0%	491,699	0	2.668e+09	3.642e+04	3.982e+06	0
EXTOT	1,070,994	100.0%	432,572	0	4.668e+09	9.119e+04	6.508e+06	0
BLDFRONT	1,070,994	100.0%	228,815	0	7.575e+03	2.304e+01	3.558e+01	0
BLDDEPTH	1,070,994	100.0%	228,853	0	9.393e+03	3.992e+01	4.271e+01	0
AVLAND2	282,726	26.4%	0	3	2.371e+09	2.462e+05	6.179e+06	2,408
AVTOT2	282,732	26.4%	0	3	4.501e+09	7.139e+05	1.165e+07	750
EXLAND2	87,449	8.2%	0	1	2.371e+09	3.512e+05	1.080e+07	2,090
EXTOT2	130,828	12.2%	0	7	4.501e+09	6.568e+05	1.607e+07	2,090

2. Categorical Table

Field Name	# records with values	% Populated	# Zeros	# Unique Values	Most Common
RECORD	1,070,994	100.0%	0	1,070,994	1
BBLE	1,070,994	100.0%	0	1,070,994	1000010101
BORO	1,070,994	100.0%	0	5	4
BLOCK	1,070,994	100.0%	0	13,984	3944
LOT	1,070,994	100.0%	0	6,366	1
EASEMENT	4,636	0.4%	0	12	E
OWNER	1,039,249	97.0%	0	863,347	PARKCHESTER PRESERVAT
BLDGCL	1,070,994	100.0%	0	200	R4
TAXCLASS	1,070,994	100.0%	0	11	1
EXT	354,305	33.1%	0	3	G
EXCD1	638,488	59.6%	0	129	1017
STADDR	1,070,318	99.9%	0	839,280	501 SURF AVENUE
ZIP	1,041,104	97.2%	0	196	10314
EXMPTCL	15,579	1.5%	0	14	X1
EXCD2	92,948	8.7%	0	60	1017
PERIOD	1,070,994	100.0%	0	1	FINAL
YEAR	1,070,994	100.0%	0	1	2010/11
VALTYPE	1,070,994	100.0%	0	1	AC-TR

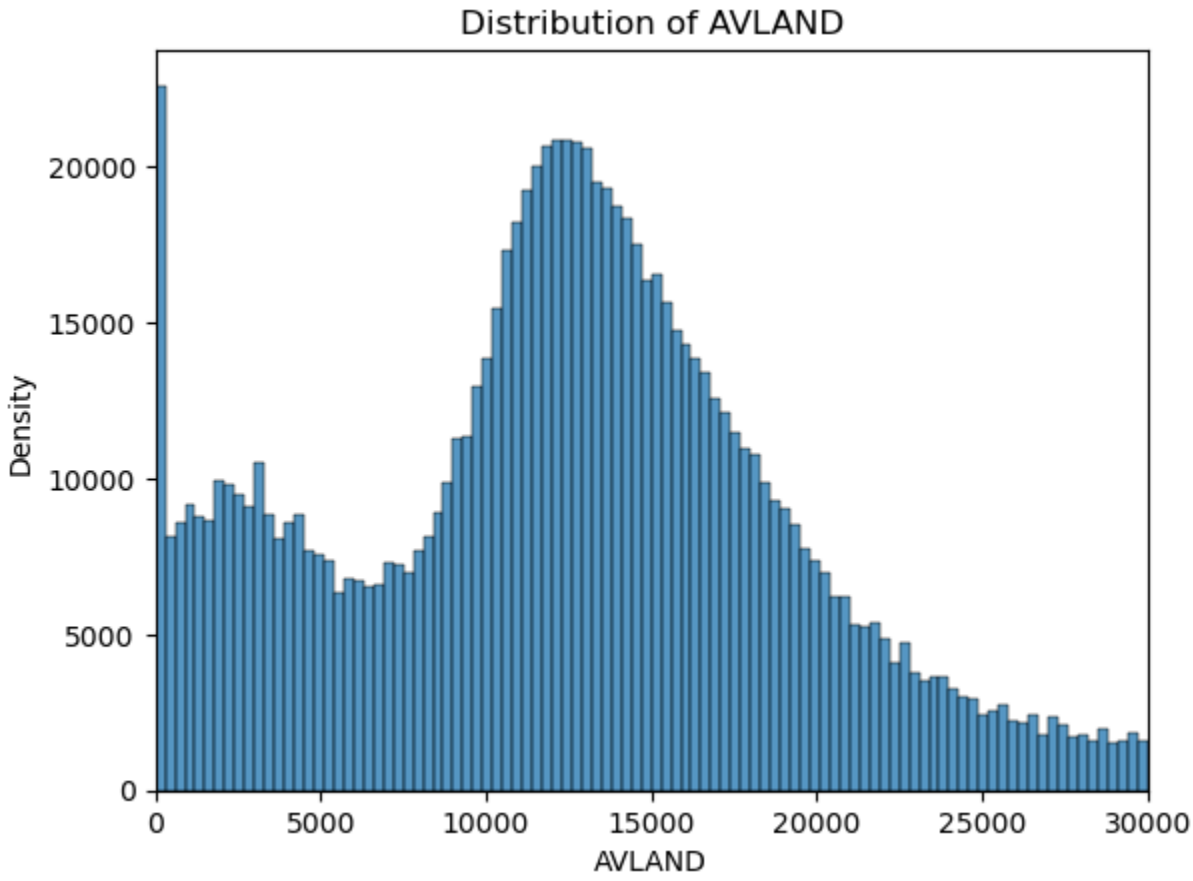
Important Field Distributions

1. **Field Name: FULLVAL**



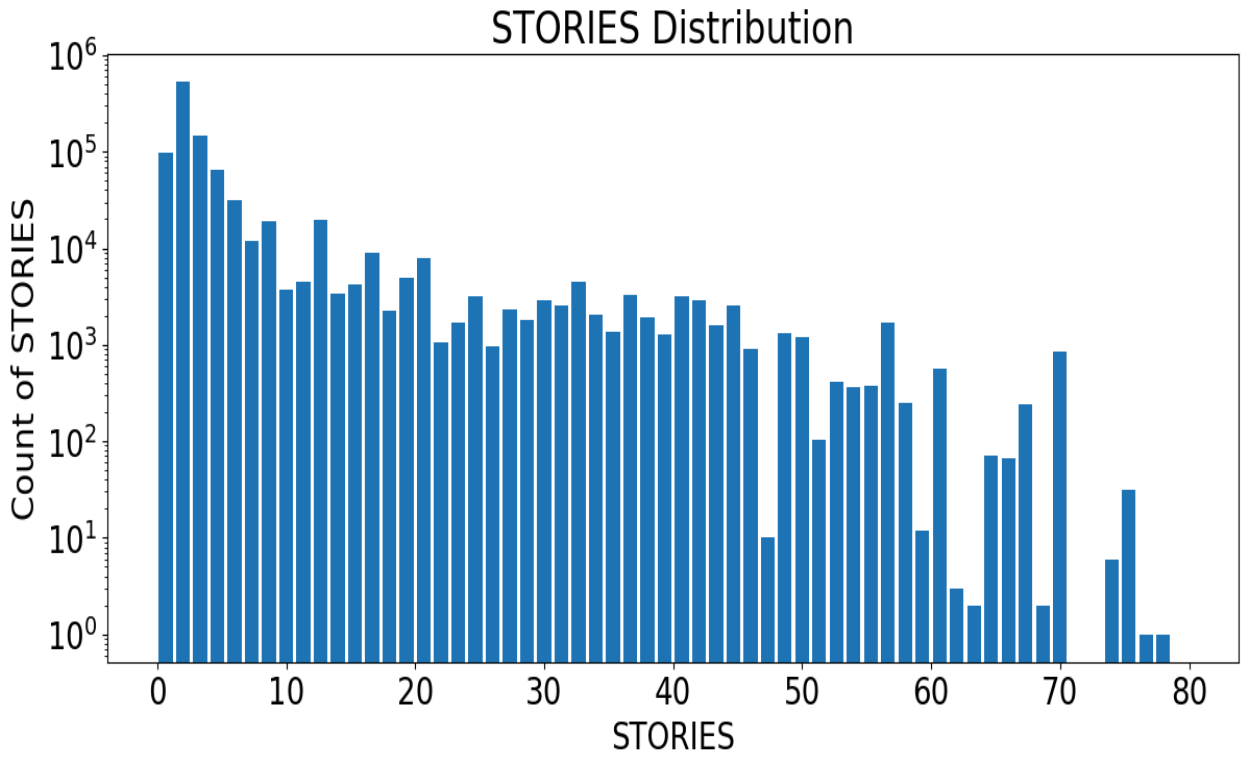
The distribution of the FULLVAL field reveals a bimodal pattern, with a significant number of small values concentrated near zero and a second peak around 400,000. The density gradually declines as values approach the upper limit of 1,000,000. This distribution highlights variability in property values, with most properties falling below 500,000.

2. **Field Name: AVLAND**



The distribution of AVLND exhibits a clear peak around 10,000, indicating the most common land values in the dataset. The density decreases gradually beyond this point, with fewer occurrences approaching the upper limit of 30,000. The data highlights a concentration of smaller values, with a visible skewness toward lower land values.

3. **Field Name: STORIES**



The STORIES field represents the number of floors in a building. The distribution is right-skewed, with most buildings having fewer than 10 stories. A small number of properties have more than 50 stories, with the tallest buildings reaching 80 stories. The logarithmic scale highlights the dominance of low-rise structures in the dataset.

Section 3. Data Cleaning

1. Outliers

Outliers are not removed from this dataset. Outliers are not removed from this dataset. Retaining outliers is critical as they may represent significant anomalies or unusual patterns relevant to detecting property tax fraud. These data points could provide valuable insights rather than being dismissed as noise.

2. Exclusions

For this project, we identified and excluded records that were not relevant to detecting property tax fraud. The exclusions were applied to ensure the dataset focuses on privately-owned properties, as fraud analysis for government or public properties is outside the scope. Below are the specific exclusion criteria:

1. **Easement Type**:

   Why: Properties with specific easement types (e.g., government or undefined usage) may not be relevant for fraud detection since they often belong to public entities or organizations exempt from typical property tax regulations.

   What: Records where the EASEMENT field did not equal "U" (unspecified or irrelevant easement type) were excluded. This resulted in the removal of 1 record.

2. **Government or Cemetery Owners**:

   Why: Properties owned by government entities, public organizations, or cemeteries are not the focus of this fraud analysis, as their property characteristics and tax obligations differ significantly from privately-owned properties.

   What: A custom list (gov_list) of terms indicative of government or cemetery ownership (e.g., “UNITED STATES,” “GOVERNMENT,” “CEMETERY”) was created to identify and exclude records. Any matching government or organizational entities were added to the exclusion list (remove_list). Examples include “NYC DEPT OF HIGHWAYS” and “MOUNT HOPE CEMETERY.”

3. **Frequent Owners**:

   Why: Frequently occurring organizational names, such as city housing authorities or public entities, were identified as likely irrelevant for fraud detection. These entities often

represent large portfolios of properties managed by government agencies or non-profit organizations.

What: The top 20 most frequent owner names were analyzed, and additional names indicative of public or organizational ownership were added to a second exclusion list (remove_list2). Examples include "PARKCHESTER PRESERVATION", "NYC HOUSING PARTNERSHIP", and "CITY OF NEW YORK". Specific entities that were determined to be relevant (e.g., “YORKVILLE TOWERS ASSO”, “434 M LLC”) were removed from the exclusion list to retain private property owners. These exclusions were made to refine the dataset and focus on private ownership relevant to fraud detection.

A total of 26,502 records were excluded from the dataset based on the above criteria, primarily focusing on public, government, and organizational ownerships irrelevant to the fraud analysis.

3. Imputation

To ensure the completeness and usability of the dataset, missing values in key fields were imputed using logical and systematic approaches. Below is the step-by-step description of the imputation process:

1. ZIP Code Imputation Process

Why: ZIP codes are critical for geographic and fraud-related analyses. Missing ZIP codes can lead to inaccuracies in downstream tasks, reducing the quality and reliability of the dataset.

What and How:

A total of 20,431 records (approximately 2% of the data) were missing ZIP codes.

Step 1: Filled missing ZIP codes by mapping them using a combination of street address (STADDR) and borough (BORO). This resolved 2,832 records, leaving 17,599 missing ZIP codes.

Step 2: Observed that the data appeared to be sorted by ZIP codes due to how it was collected and stored. Using this, missing ZIP codes were filled based on the values of surrounding records:

If the ZIP codes for both the previous and next records were the same, the missing value was replaced by that ZIP code. This resolved 16,126 records, leaving 1,473 still missing.

Step 3: For the remaining missing ZIP codes, the values were filled using the ZIP code from the record directly above.

After these steps, all missing ZIP codes were imputed, resulting in 0 missing ZIP codes in the dataset.

2. FULLVAL Imputation Process

Why: Property values are essential for analyzing potential fraud. A value of zero or missing is treated as invalid, as all properties should have a reasonable valuation. Ensuring accurate values for FULLVAL improves the detection of unusual or fraudulent property assessments.

What and How:

Step 1: We have 10,025 records with missing or zero values in the FULLVAL field, treating zeros as missing since all properties should have a value.

Step 2: We group all records by the combination of TAXCLASS, BORO, and BLDGCL, calculate the average of FULLVAL for each group, and replace the missing values in that group with the group average. This step fills in 2,718 records, leaving 7,307 remaining.

Step 3: We then group all remaining records by the combination of TAXCLASS and BORO, calculate the average of FULLVAL for each group, and replace the missing values in that group with the group average. This step fills in 6,921 records, leaving 386 remaining.

Step 4: We then group all remaining records by TAXCLASS, calculate the average of FULLVAL for each group, and replace the missing values in that group with the group average. This step fills in all remaining 386 records, resulting in no missing or zero values for FULLVAL.

3. AVLAND Imputation Process

Why: These field represents the assessed value of land (AVLAND). Missing or zero values could indicate either data entry errors or cases needing further scrutiny. Accurate imputation ensures these fields provide meaningful data for identifying irregularities in property assessments.

What and How:

Step 1: We identify 10,027 records with the AVLAND field either missing or having a value of zero. Zeros are treated as missing since all properties should have a value. These records are flagged for imputation.

Step 2: We group all records by the combination of TAXCLASS, BORO, and BLDGCL, calculate the average of AVLAND for each group, and replace the missing values in that group with the average of the group. This step fills in 2,718 records, leaving 7,307 records still missing.

Step 3: We then group all records by the combination of TAXCLASS and BORO, calculate the average of AVLAND for each group, and replace the missing values in that group with the average of the group. This step fills in 6,923 records, leaving 386 records still missing.

Step 4: Finally, we group all records by TAXCLASS, calculate the average of AVLAND for each group, and replace the missing values in that group with the average of the group. This step fills in the remaining 386 records, resulting in all records now having a reasonable value for the AVLAND field.

4. AVTOT Imputation Process

Why: These field represents the assessed value of land (AVLAND) and the total assessed value (AVTOT). Missing or zero values could indicate either data entry errors or cases needing further scrutiny. Accurate imputation ensures these fields provide meaningful data for identifying irregularities in property assessments.

What and How:

Step 1: We identify 10,025 records with the AVTOT field either missing or having a value of zero. Zeros are treated as missing since all properties should have a value. These records are flagged for imputation.

Step 2: We group all records by the combination of TAXCLASS, BORO, and BLDGCL, calculate the average of AVTOT for each group, and replace the missing values in that group with the average of the group. This step fills in 2,718 records, leaving 7,307 records still missing.

Step 3: We then group all records by the combination of TAXCLASS and BORO, calculate the average of AVTOT for each group, and replace the missing values in that group with the average of the group. This step fills in 6,921 records, leaving 386 records still missing.

Step 4: Finally, we group all records by TAXCLASS, calculate the average of AVTOT for each group, and replace the missing values in that group with the average of the group. This step fills in the remaining 386 records, resulting in all records now having a reasonable value for the AVTOT field.

5. **STORIES Imputation Process**

Why: The number of STORIES in a building is an important characteristic affecting property valuation. Missing data here could misrepresent property characteristics, leading to faulty fraud detection models.

What and How:

Step 1: We identify 42,029 records with the STORIES field missing. These records are flagged for imputation since STORIES is an important variable for analysis.

Step 2: We group all records by the combination of BORO and BLDGCL, calculate the mode of STORIES for each group, and replace the missing values in that group with the mode. This step fills in 4,108 records, leaving 37,921 records still missing.

Step 3: We then group all records by TAXCLASS, calculate the average of STORIES for each group, and replace the remaining missing values in that group with the mean. This step fills in the remaining 37,921 records, resulting in no missing values in the STORIES field.

6. **LTFRONT Imputation Process**

Why: These fields represent the frontage and depth of a lot, which are fundamental for determining property dimensions. Missing or invalid values (e.g., 0 or 1) hinder accurate analysis and comparisons of property characteristics, which are crucial for fraud detection.

What and How:

Step 1: We began by identifying records in the LTFRONT field where values were either 0 or 1. These values are treated as invalid because we assume that all properties should have valid frontage measurements. A total of 161,133 records were identified with missing values in the LTFRONT field.

Step 2: The dataset was grouped by TAXCLASS, BORO, and BLDGCL, and the mean LTFRONT value was calculated for each group. Missing values in the LTFRONT field were then replaced with the mean value of their respective group. After this step, a significant portion of missing values was filled.

Step 3: For the remaining missing values, the dataset was grouped only by TAXCLASS, and the mean LTFRONT value was calculated for each class. Missing values were replaced with the mean value of their respective TAXCLASS. This process resolved all missing values, leaving the LTFRONT field fully imputed with reasonable estimates.

7. LTDEPTH Imputation Process

Why: These fields represent the frontage and depth of a lot, which are fundamental for determining property dimensions. Missing or invalid values (e.g., 0 or 1) hinder accurate analysis and comparisons of property characteristics, which are crucial for fraud detection.

What and How:

Step 1: The LTDEPTH field was analyzed to identify missing or invalid values (e.g., 0 or 1). A total of 161,715 records were identified as having missing or invalid values, which we treated as needing imputation since all properties should have a valid lot depth.

Step 2: The dataset was grouped by TAXCLASS, BORO, and BLDGCL, and the mean LTDEPTH value was calculated for each group. Missing values in the LTDEPTH field were replaced with the mean value from their respective group. This step significantly reduced the number of missing values.

Step 3: The remaining missing values were imputed by grouping the dataset by TAXCLASS alone. The mean LTDEPTH value was calculated for each tax class, and missing values were replaced with these averages. After this step, all missing values in the LTDEPTH field were resolved.

8. BLDFRONT Imputation Process

Why: These represent the frontage and depth of a building. Accurate dimensions are vital for evaluating whether property characteristics align with reported tax assessments. Missing or invalid data compromises the integrity of such evaluations.

What and How:
Step 1: Identify missing or invalid values in the BLDFRONT field. Initially, count the missing or invalid values (e.g., 0 or 1) and replace these invalid values with NaN to standardize the representation of missing data.

Step 2: Group the data by TAXCLASS, BORO, and BLDGCL. For each unique combination of these fields, calculate the mean BLDFRONT. Replace the missing values within each group with the calculated group mean. This imputation ensures that the values reflect specific property characteristics.

Step 3: Check for any remaining missing values. If missing values still exist, group the data by TAXCLASS and BORO only. Calculate the mean BLDFRONT for each group and use these values to fill in the remaining missing entries.

Step 4: Perform a final imputation step by grouping the data solely by TAXCLASS. Calculate the mean BLDFRONT for each tax class and use these values to fill in any remaining missing data. This ensures that no missing values are left.

Step 5: Verify that all missing values in the BLDFRONT field have been addressed. A final check confirms that the count of missing values is now zero, indicating that the imputation process is complete and the data is ready for further analysis.

9. BLDDEPTH. Imputation Process

Why: These represent the frontage and depth of a building. Accurate dimensions are vital for evaluating whether property characteristics align with reported tax assessments. Missing or invalid data compromises the integrity of such evaluations.

What and How:

Step 1: Identified missing or invalid values in BLDFRONT. Treated zeros and ones as invalid.

Step 2: Grouped records by TAXCLASS, BORO, and BLDGCL to calculate the group mean for BLDFRONT. Replaced missing values with the group mean, resolving most cases.

Step 3: Grouped the remaining records by TAXCLASS and BORO to calculate the group mean. Replaced missing values with the mean, resolving nearly all cases.

Step 4: Grouped the remaining records by TAXCLASS and replaced missing values with the TAXCLASS mean. This step resolved all missing BLDFRONT values.

Section 4. Description of the Variable

Description of Variables	# Variables Created
Entities: This dataset includes 32 variables representing property identifiers (e.g., RECORD, BBLE, BORO, BLOCK, LOT), physical attributes (e.g., LTFRONT, LTDEPTH, STORIES, BLDFRONT, BLDDEPTH), valuation metrics (e.g., FULLVAL, AVLAND, AVTOT, EXLAND, EXTOT), classification codes (e.g., TAXCLASS, EXMPTCL), exemption details (e.g., EXLAND2, AVLAND2), and descriptive information (e.g., OWNER, ZIP, YEAR), collectively capturing property identification, valuation, and tax-related details.	32
Lot Size (ltsize): Product of LTFRONT and LTDEPTH plus epsilon to avoid division by zero. Used to calculate size of the land.	1
Building Size (bldsize): Product of BLDFRONT and BLDDEPTH plus epsilon. Measures the building footprint.	1
Building Volume (bldvol): Product of bldsize and STORIES plus epsilon. Represents the total building volume.	1
FULLVAL Ratios (r1, r2, r3): Ratios of full value (FULLVAL) divided by land size (ltsize), building size (bldsize), and building volume (bldvol). Used to detect overvaluation or undervaluation.	3
AVLAND Ratios (r4, r5, r6): Ratios of assessed land value (AVLAND) divided by ltsize, bldsize, and bldvol. Highlights discrepancies in assessed land valuation.	3
AVTOT Ratios (r7, r8, r9): Ratios of total assessed value (AVTOT) divided by ltsize, bldsize, and bldvol. Detects anomalies in total property assessment.	3
Inverse Ratios: r1inv, r2inv, r3inv, r4inv, r5inv, r6inv, r7inv, r8inv, r9inv are the inverses of the primary ratios (r1 to r9). These capture extreme values, particularly very low ratios, that might be overlooked by the primary variables alone.	9
Standardized Ratios by ZIP (r1_zip5 to r9_zip5): Ratios (r1 to r9) standardized by the average for each ZIP group. These capture anomalies specific to neighborhoods	9
Standardized Ratios by TAXCLASS (r1_taxclass to r9_taxclass): Ratios (r1 to r9) standardized by the average for each TAXCLASS. These highlight outliers within tax classifications.	9

value_ratio: Ratio of FULLVAL to the sum of AVLAND and AVTOT. Captures discrepancies between full property value and assessed values.	1
size_ratio: Ratio of bldsize to ltsize. Identifies anomalies in building-to-land size proportions.	1

Motivation Behind the Variables

The variables designed for this fraud detection framework are motivated by the need to capture anomalies in property valuations. Each variable group serves a specific purpose in identifying unusual records, leveraging both physical characteristics and financial assessments to uncover potential discrepancies.

The lot size, building size, and building volume variables represent the physical dimensions of a property, including the land area, building footprint, and total building volume. These foundational metrics are crucial for scaling valuation data. Anomalies in these dimensions may reveal discrepancies, such as a small lot with an unusually high assessed value indicating potential overvaluation or manipulation. Similarly, a large lot with an abnormally low valuation may suggest intentional underreporting for tax avoidance.

The primary ratios (r1 to r9) are created by dividing key valuation metrics (FULLVAL, AVLAND, AVTOT) by the size variables (ltsize, bldsize, bldvol). These ratios standardize valuation data by physical dimensions, enabling comparisons across properties of different sizes. Outliers in these ratios often signal irregularities, such as a high FULLVAL/ltsize ratio pointing to potential overvaluation or a low AVLAND/bldsize ratio indicating under-assessment of the land portion.

To ensure both extremes are captured, inverse ratios (r1inv to r9inv) are calculated as the maximum of the original ratio and its inverse. This adjustment emphasizes both unusually high and low values, which are often indicative of anomalies. For instance, an excessively high ratio could reflect an inflated property valuation to secure larger loans, whereas a very low ratio might point to undervaluation for reducing taxes or fees.

The grouped averages by ZIP and TAXCLASS provide local context by standardizing property ratios relative to their geographic and tax classification peers. This approach highlights properties that deviate significantly from neighborhood or group norms. For example, a property with a much higher FULLVAL/ltsize ratio than the ZIP code average may indicate misreporting or fraudulent appraisals specific to that area. Similarly, a low AVTOT/bldvol ratio within a TAXCLASS group could reveal undervaluation to minimize tax liabilities.

Finally, the additional ratios, including the value ratio (value_ratio) and size ratio (size_ratio), focus on discrepancies between valuations and physical dimensions. The value_ratio compares the full valuation (FULLVAL) to the combined assessed values of land and total (AVLAND + AVTOT), with deviations suggesting inconsistencies between market and official assessments. The size_ratio compares building size (bldsize) to lot size (ltsize), with extreme values possibly pointing to misreported construction or undeclared property modifications.

Types of Anomalies Being Targeted

The variables aim to detect several key types of anomalies in property data, including overvaluation, undervaluation, and misreported property characteristics. Overvaluation is identified through high ratios, such as a disproportionately large FULLVAL/ltsize, which might indicate inflated property values for purposes such as securing larger loans or avoiding tax scrutiny. Conversely, undervaluation is detected through low ratios, such as a small AVLAND/bldsize, which could suggest intentional underreporting of property value to minimize tax liabilities.

Misreported property characteristics are another type of anomaly targeted by the analysis. These discrepancies can include misalignments between physical attributes, such as a size_ratio that indicates undeclared construction or an incorrect reporting of building size relative to lot size. Contextual anomalies are also a focus, as properties that deviate significantly from their neighborhood averages (e.g., ZIP-based ratios) or tax classification norms may highlight systemic issues, errors, or fraudulent reporting. By focusing on these specific anomalies, the framework provides actionable insights into irregularities in property valuation data.

Section 5. Dimensionality Reduction

Description of What, Why, and How

Dimensionality reduction simplifies high-dimensional data by reducing the number of variables while retaining the most relevant information. In this project, it ensures that property valuation data, which includes numerous correlated and derived variables, is optimized for anomaly detection. This process eliminates redundancy, reduces computational complexity, and enhances the clarity of the analysis.

Dimensionality reduction is crucial because high-dimensional datasets often include noise and irrelevant information. By focusing on the most significant patterns in the data, this technique preserves the variance and relationships critical for detecting anomalies. In this project, Principal Component Analysis (PCA) was used after z-scaling the data to achieve dimensionality reduction. PCA transforms the original dataset into uncorrelated principal components, with only the most informative components retained.

Description of Z-Scaling

What it is:

Z-scaling, or standardization, is a transformation process that adjusts data so that each variable has a mean of 0 and a standard deviation of 1. This ensures that variables with different units or scales are made comparable.

Why it is necessary:

Z-scaling is essential in fraud detection because many property variables, such as monetary values (e.g., FULLVAL, AVLAND) and physical dimensions (e.g., LTFRONT, BLDFRONT), vary widely in magnitude. Without standardization, larger variables might dominate the analysis, skewing the results. Z-scaling ensures all variables contribute equally to the detection algorithms.

How it is done:

For each variable x , z-scaling is applied using the formula:

$$Z = \frac{x - \mu}{\sigma}$$

where \mu is the mean of the variable and \sigma is its standard deviation. After z-scaling, the dataset is balanced, allowing both Principal Component Analysis (PCA) and anomaly detection algorithms to work effectively.

Description of PCA (Principal Component Analysis)

What it is:
PCA is a dimensionality reduction technique that transforms a high-dimensional dataset into a smaller set of uncorrelated variables called principal components. These components capture the majority of the dataset's variance, making the data easier to analyze.
Why it is necessary:
In high-dimensional datasets like this one, many variables are correlated, leading to redundancy and noise. PCA eliminates these redundancies while retaining the most informative features, simplifying the dataset and reducing computational complexity.

How it is done:

1. **Calculate the Covariance Matrix**: PCA starts by computing the covariance matrix of the standardized dataset.

2. **Compute Eigenvalues and Eigenvectors**: The eigenvalues represent the variance explained by each component, and the eigenvectors define the principal component directions.

3. **Transform the Data**: The original data is projected onto the principal component axes, creating a reduced dataset. Typically, only the top components (e.g., the first 5) that explain the majority of the variance are retained.

PCA provides a transformed dataset with fewer variables, which are z-scaled again to ensure balanced importance during anomaly detection. This process eliminates noise and redundancy, improving the performance of subsequent fraud detection algorithms.

Section 6. Anomaly Detection Algorithms

The goal of anomaly detection in this project is to identify properties with unusual patterns in valuation, dimensions, or classification by using two scoring methods: **Minkowski Distance-Based Scoring** and **Autoencoder Reconstruction Error Scoring**. These methods analyze the reduced dataset (transformed by dimensionality reduction) and produce separate scores that are subsequently scaled and combined to rank anomalies.

The Minkowski distance method calculates the "distance" of each record from the center (mean) of the dataset in the reduced PCA space. The larger the distance, the more anomalous the property is considered to be.
Equation:
The Minkowski distance is defined as:

$$D(\mathbf{x}, \boldsymbol{\mu}) = \left(\sum_{i=1}^{n} |x_i - \mu_i|^p \right)^{\frac{1}{p}}$$

Where:

- \mathbf{x} is the data point (property) in the reduced PCA space.
- \mathbf{\mu} is the mean vector of the dataset in the reduced PCA space.
- n is the number of dimensions (principal components).
- p is the Minkowski parameter (commonly p = 2 for Euclidean distance or p = 1 for Manhattan distance).

Steps:

1. **Calculate the Mean Vector**:

   Compute the mean (\mu) of all records in each principal component dimension.

2. **Compute Distances**:

   For each property, calculate the Minkowski distance from the mean.

3. **Scale the Scores**:

Normalize the distances using z-scaling to create the Minkowski-based anomaly score.

Why It Works:
Properties far from the center in PCA space deviate significantly from the dataset's overall patterns, indicating potential anomalies.

Autoencoder Reconstruction Error Scoring

What It Does:
An autoencoder is a neural network that learns to compress and reconstruct input data. Anomalies are identified as properties with high reconstruction errors, as they deviate from the typical patterns learned by the autoencoder.

Reconstruction Error Equation:
The reconstruction error for a data point `\mathbf{x}` is calculated as:

$$E(\mathbf{x}) = \sum_{i=1}^{n} (x_i - \hat{x}_i)^2$$

Where:

- `\mathbf{x}` is the original data point in the reduced PCA space.
- `\hat{\mathbf{x}}` is the reconstructed data point (output of the autoencoder).
- `n` is the number of dimensions (principal components).

Steps:

1. **Train the Autoencoder**:

   Use the reduced dataset to train an autoencoder with a bottleneck layer smaller than the input size. The bottleneck layer forces the network to compress the data, learning its essential patterns.

2. **Reconstruct Data**:

   Pass each property through the trained autoencoder to obtain reconstructed values (`\hat{\mathbf{x}}`).

3. **Compute Reconstruction Error**:

   Calculate the reconstruction error for each property using the equation above.

4. **Scale the Scores**:

   Normalize reconstruction errors using z-scaling to create the autoencoder-based anomaly score.

Why It Works:
Properties with high reconstruction errors deviate significantly from the patterns the autoencoder has learned, indicating that they are unusual.

Combining the Scores

Why Combine Scores?
Each method captures different aspects of anomalies:

- Minkowski distance focuses on overall deviation in the reduced space.
- Autoencoder error highlights records poorly represented by the learned data structure.

By combining scores, the strengths of both methods are leveraged for more robust anomaly detection.

Steps to Scale and Combine:

1. **Scale Scores**:
   Both Minkowski and autoencoder scores are z-scaled to ensure they have comparable ranges.

   - Z-scaling equation:

$$Z = \frac{X - \mu}{\sigma}$$

Where \mu is the mean and \sigma is the standard deviation of the scores.

2. **Rank Scores**:

Rank properties based on their z-scaled scores separately for each method.

3. **Combine Ranks**:

   Average the ranks from both methods to create a final score:

$$\text{Final Score} = \frac{\text{RankMinkowski} + \text{RankAutoencoder}}{2}$$

4. **Interpretation**: Properties with the highest final scores are flagged as anomalies and prioritized for further investigation. This combined approach ensures robustness by leveraging both linear (Z-score) and non-linear (Autoencoder) perspectives.

The two-score anomaly detection system, combining Minkowski distance and autoencoder reconstruction error, provides a robust framework for identifying unusual properties. One of its key benefits is flexibility, as it integrates both linear and non-linear techniques. Minkowski distance captures overall deviations from the dataset's mean in the reduced PCA space, while the autoencoder focuses on properties poorly reconstructed by the learned data structure, highlighting anomalies that deviate from typical patterns in complex ways.

This approach is also highly robust, as it mitigates biases inherent in relying on a single detection method. By combining the strengths of both algorithms, the framework ensures that no significant anomalies are overlooked, improving the consistency and accuracy of fraud detection. Furthermore, the system is scalable, making it adaptable to datasets of varying sizes and complexities. Whether used for high-dimensional property datasets or other anomaly-prone domains, this dual-method approach ensures comprehensive coverage of unusual records, enhancing the reliability of the results.

Section 7. Results

The results of the anomaly detection framework are summarized in terms of how to interpret and use the scores, how to prioritize and examine flagged properties, and detailed descriptions of three interesting case studies from the analysis.

Using the Anomaly Score

The final anomaly score is computed by combining the Minkowski distance-based rank and autoencoder reconstruction error rank, providing a comprehensive measure of how unusual a property is. Properties with higher final scores are more likely to exhibit irregularities in their valuation, dimensions, or tax classification, making them strong candidates for further investigation

.

To use the score:

1. **Sort Properties**: Properties are ranked in descending order of their final scores.

2. **Threshold Selection**: Depending on the resources available, a threshold can be set (e.g., top 1% of scores) to focus on the most anomalous records.

3. **Investigative Prioritization**: High-scoring properties are flagged for manual review, where property records, ownership details, and historical valuations are analyzed to confirm potential irregularities.

Examining the Top Properties

Once properties are ranked by their scores, the top properties should be closely examined to identify patterns or anomalies. The following steps guide this process:

1. **Review Valuation Metrics**: Compare reported valuations (e.g., FULLVAL, AVLAND) against benchmarks for similar properties in the same ZIP or TAXCLASS.

2. **Analyze Physical Dimensions**: Verify whether reported dimensions (e.g., STORIES, LTFRONT) align with the observed characteristics from maps or satellite imagery.

3. **Identify Contextual Deviations**: Use standardized ratios (by ZIP and TAXCLASS) to understand how the property deviates from its peers in local or classification-specific contexts.

This method ensures a systematic approach to investigating anomalies and focusing on properties with the most suspicious patterns.

Examination of the Final Score

Understanding the distribution and properties of the final fraud scores is essential to interpreting the results and prioritizing investigations. Descriptive statistics provide valuable insights into how scores are distributed and identify patterns that may indicate anomalies.

The mean final score of 522,246.50 and a standard deviation of 298,308.51 reveal a significant spread in scores, highlighting variability among the properties analyzed. The minimum score of 21.50 and the maximum score of 1,044,492.00 demonstrate a wide range, reflecting both extremely low and high scoring properties. The median score of 521,594.50, closely aligning with the mean, suggests a relatively balanced distribution of scores around the central value. However, the substantial standard deviation and the range between the 25th percentile (267,732.00) and the 75th percentile (784,299.62) indicate the presence of outliers, with properties at the extremes deviating significantly from the majority.

These descriptive statistics are critical for identifying anomalies. Properties with the highest final scores—those in the top percentile—are likely to represent the most significant deviations from expected patterns, warranting further scrutiny. Such properties may exhibit characteristics indicative of tax fraud, underreporting of valuations, or other irregularities. Similarly, properties with unusually low scores may also represent anomalies, possibly due to data entry errors or unique property features that diverge from the norm.

By focusing investigative efforts on the properties with the most extreme scores, the fraud detection process can be streamlined and made more effective. This detailed examination of the distribution and variability of scores ensures that resources are directed toward the most suspicious cases, enhancing the accuracy and efficiency of fraud detection. Moreover, analyzing score distributions helps refine the detection model by highlighting patterns or systemic issues that may require algorithmic adjustments in future iterations.

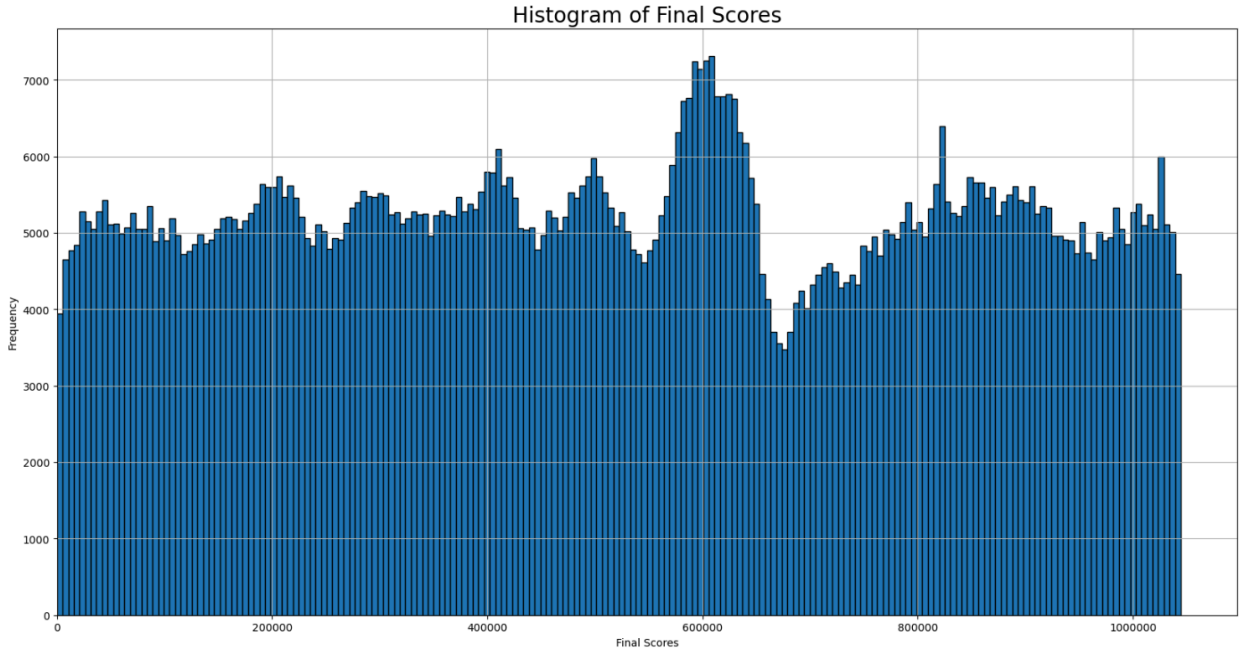
Visualization – Histogram

The histogram of final fraud scores reveals a multi-modal distribution, with several distinct peaks that indicate clustering within the data. A prominent peak is observed around the 600,000 range, with over 7,000 properties scoring in this region, highlighting a concentration of properties with similar characteristics or valuation patterns. Additional peaks are noticeable around 200,000, 400,000, 800,000, and 1,000,000, suggesting that properties tend to fall within specific score

bands. These bands may correspond to distinct property types, valuation tiers, or systematic scoring criteria, reflecting natural groupings within the dataset.

The scores span a wide range, from 0 to 1,050,000, showcasing the diversity of properties analyzed. Unlike a typical skewed distribution, the scores are relatively evenly spread across the range, with no single dominant cluster and minimal skewness toward the lower or upper extremes. The presence of multiple peaks further suggests the existence of sub-groups or property classifications within the data, potentially linked to tax classes, geographic areas, or valuation metrics.

Interestingly, certain higher-frequency spikes at specific score intervals stand out from the otherwise smooth distribution. These anomalies may represent systematic scoring patterns, data irregularities, or potentially fraudulent activities. Such deviations warrant closer inspection to determine whether they reflect legitimate property groupings or misreported characteristics. By analyzing these patterns and anomalies, investigators can refine their understanding of the dataset and prioritize records that merit further examination. The histogram thus provides a visual foundation for exploring scoring patterns and guiding targeted investigations into suspicious properties.

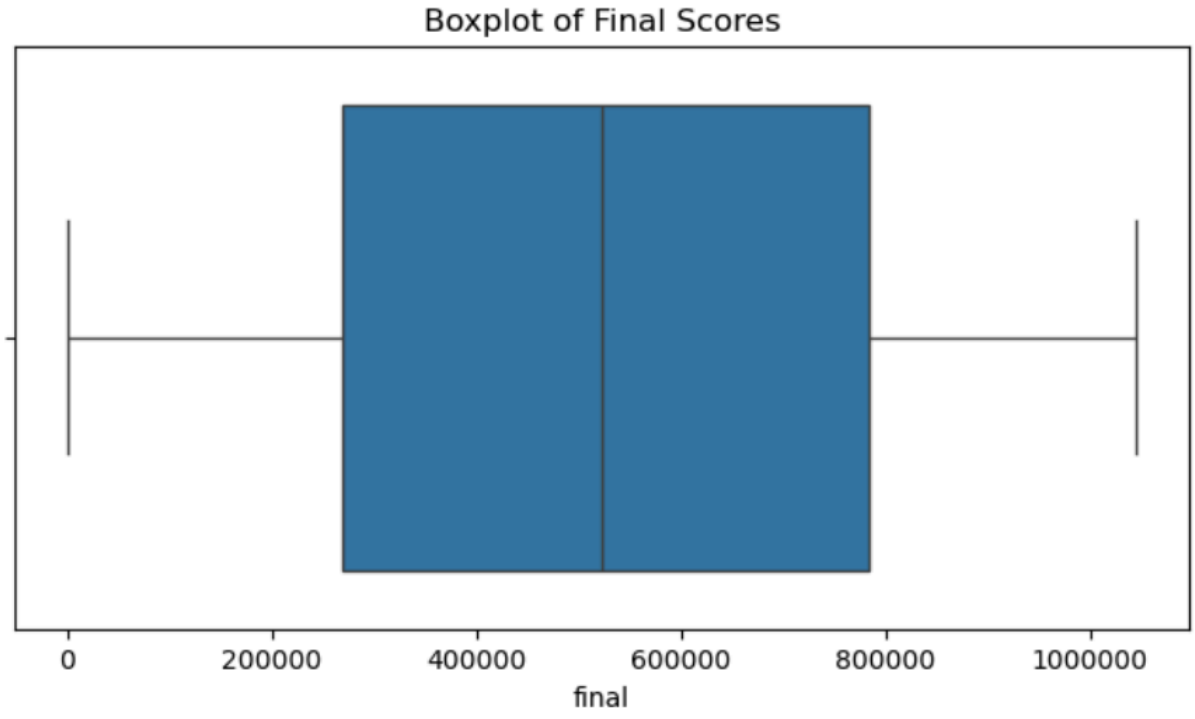


Visualization – Boxplot

The boxplot provides a comprehensive visual summary of the distribution of final fraud scores, highlighting central tendencies and variability across the dataset. The **median score**, depicted by the central line within the box, is approximately **500,000**, indicating that half of the properties have scores above and half below this value. The edges of the box correspond to the **25th percentile (250,000)** and the **75th percentile (780,000)**, defining the interquartile range (IQR) of **530,000**. This IQR captures the middle 50% of the data, reflecting the typical range of scores and offering insights into the variability among the majority of property records.

The whiskers of the boxplot extend to the dataset's minimum and maximum scores, with the minimum close to **0** and the maximum exceeding **1,000,000**, illustrating the full range of values. Interestingly, there are no outliers visible in this boxplot, which suggests that all scores fall within **1.5 times the IQR** from the quartiles. This absence of extreme values highlights the consistency and balance of the scoring system, ensuring that no disproportionately high or low scores skew the analysis.

Overall, the boxplot illustrates a dataset with moderate variability but without extreme deviations. This balanced distribution reinforces the reliability of the fraud detection model and provides a strong basis for identifying suspicious records. By focusing on scores within the broader ranges of variability, investigators can confidently target anomalies without concerns about misleading or artificially extreme values.

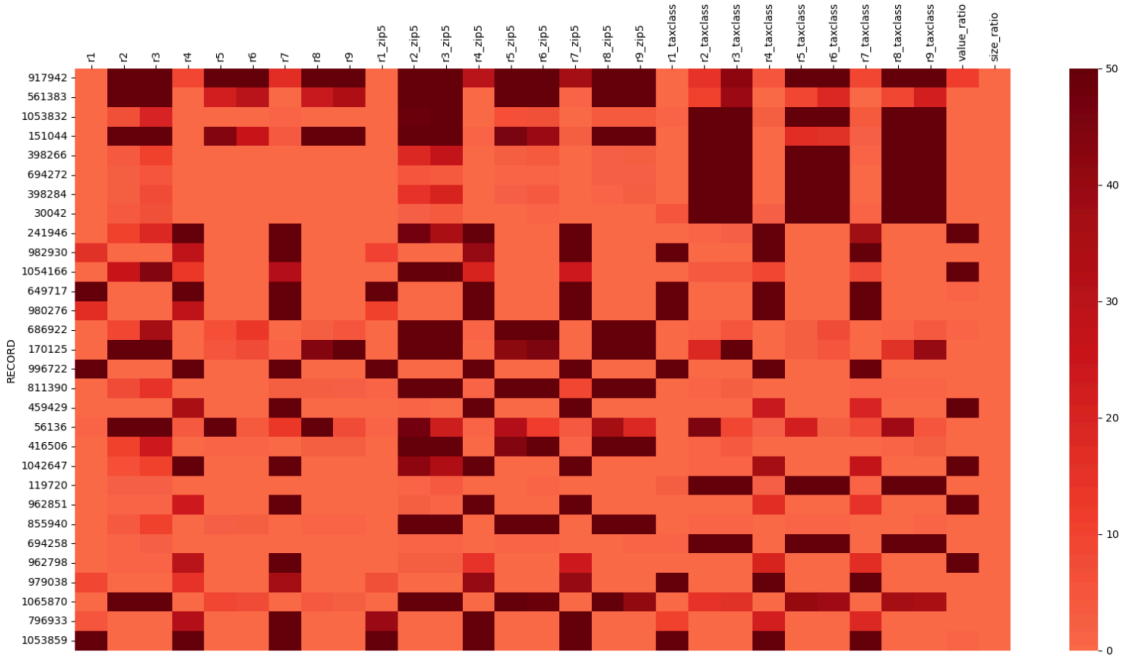


Correlation Analysis – Heatmap

The heatmap analysis provides a visual representation of the correlations between key variables and fraud scores, highlighting those most strongly associated with anomalies. Variables such as **r2**, **r3**, **r5**, **r6**, **r8**, and **r9**, along with their corresponding **zip5** and **taxclass** counterparts, consistently display darker shades, indicating stronger correlations. These patterns are particularly pronounced among the top 10 records with the highest fraud scores, underscoring their significance in identifying anomalies. Core variables like **r2**, **r3**, and **r5** likely represent fundamental metrics, such as transaction frequencies or valuation ratios, while their **zip5** counterparts reveal the geographic localization of potential irregularities.

The importance of **taxclass** variables highlights potential systemic issues or patterns related to specific tax classifications or brackets, suggesting avenues for detecting tax fraud or misuse tied to certain property types or regions. Similarly, the **value_ratio** emerges as a key metric, capturing deviations from typical valuation behaviors. Its prominence, coupled with the importance of geographic and tax-related features, suggests that anomalies often result from the interaction between transaction characteristics, regional patterns, and tax classifications.

This layered analysis highlights the need for a multifaceted approach to fraud detection. By combining transaction-based metrics, geographic information, and tax-related features, the framework achieves a robust and nuanced understanding of fraudulent activity. The interplay between these variables enhances the detection model, ensuring that anomalies arising from various dimensions are effectively identified and prioritized for investigation.



Three Cases for Further Investigation

After conducting the primary analysis, it became evident that specific records exhibit unusual or inconsistent data warranting further examination. Rather than disregarding these anomalies, the approach involves scrutinizing them to identify instances where property owners may have misrepresented their property's characteristics. Such misrepresentations have significant financial implications, as they can lead to unjustly low tax assessments. This not only reduces city revenue but also shifts the tax burden onto honest property owners who accurately report their property details. Three key categories of anomalies stand out as priorities for further investigation:

1. Unusual Building Dimensions

What We Found: Some records report building dimensions (calculated as the product of BLDFRONT and BLDDEPTH) as zero, despite the property being listed as having one or more stories (STORIES).

Why It Matters: A building with zero dimensions is logically impossible if it is reported to have stories. This inconsistency suggests that the building's footprint—either the front width (BLDFRONT), the depth (BLDDEPTH), or both—has been misreported. If building size is understated, the assessed value could be artificially reduced, lowering the owner's tax liability. Investigating such cases is crucial to ensure accurate property assessments.

2. Unusual Lot Dimensions

What We Found: Some properties record lot dimensions (LTFRONT x LTDEPTH) as zero or list either frontage (LTFRONT) or depth (LTDEPTH) as unusually small, such as less than three feet.

Why It Matters: Unrealistically small or zero lot dimensions contradict standard property records. Downplaying lot size can significantly reduce the assessed land value, leading to a lower overall property tax assessment. Whether caused by data entry errors or deliberate misrepresentation, these anomalies require closer examination to ensure accurate land valuations and equitable tax collection.

3. Unusual Value

What We Found: Certain properties report monetary values (e.g., FULLVAL, AVLAND, AVTOT) that are unrealistically low—below nominal thresholds such as $100.

Why It Matters: Extremely low property values are a red flag for potential tax fraud. If valuations are grossly understated, property owners may be avoiding their fair share of taxes. Such inaccuracies undermine the city's ability to collect rightful revenue and compromise fairness within the tax system. Addressing these discrepancies ensures that all property owners are contributing appropriately to public resources.

Case Studies of Top Properties

Case Study 1: **Property 694272 (Located at Powells Cove Blvd)**



Property Details:

- **Record ID**: 694272
- **Owner**: TSUI, TIT HOW
- **Address**: Powells Cove Blvd, ZIP 11356
- **Reported Data**:
    - **FULLVAL**: $7,597,000
    - **AVLAND**: $125,400
    - **AVTOT**: $455,820
    - **Reported Stories**: **3**

Visual Analysis:

1. **Building Structure**:
    - The property appears to have **two visible floors** with a pitched roof area.
    - While the roof may provide attic space, it does not qualify as a full third story based on standard residential classifications.
    - The surrounding properties show similar two-story structures, further supporting this observation.
2. **Potential Misclassification**:
    - The roof space may have been incorrectly included as a full story in the data, which inflates the reported "stories" value to **3**.
    - This discrepancy impacts the calculation of property ratios and valuation metrics, such as r3_taxclass and r9_taxclass, which are flagged as unusually high.

Discrepancies in the Data:

1. **Inaccurate Stories Count**:
    - The reported **3 stories** does not match the visual evidence of **2 stories** with potential attic space.
    - Overstating the stories could artificially increase valuation metrics.

2. **Valuation Anomalies**:

   - FULLVAL ($7,597,000) appears unusually high for a residential property in this area with the observed structure.
   - AVLAND and AVTOT values are disproportionately low compared to FULLVAL, further indicating potential data or reporting errors.

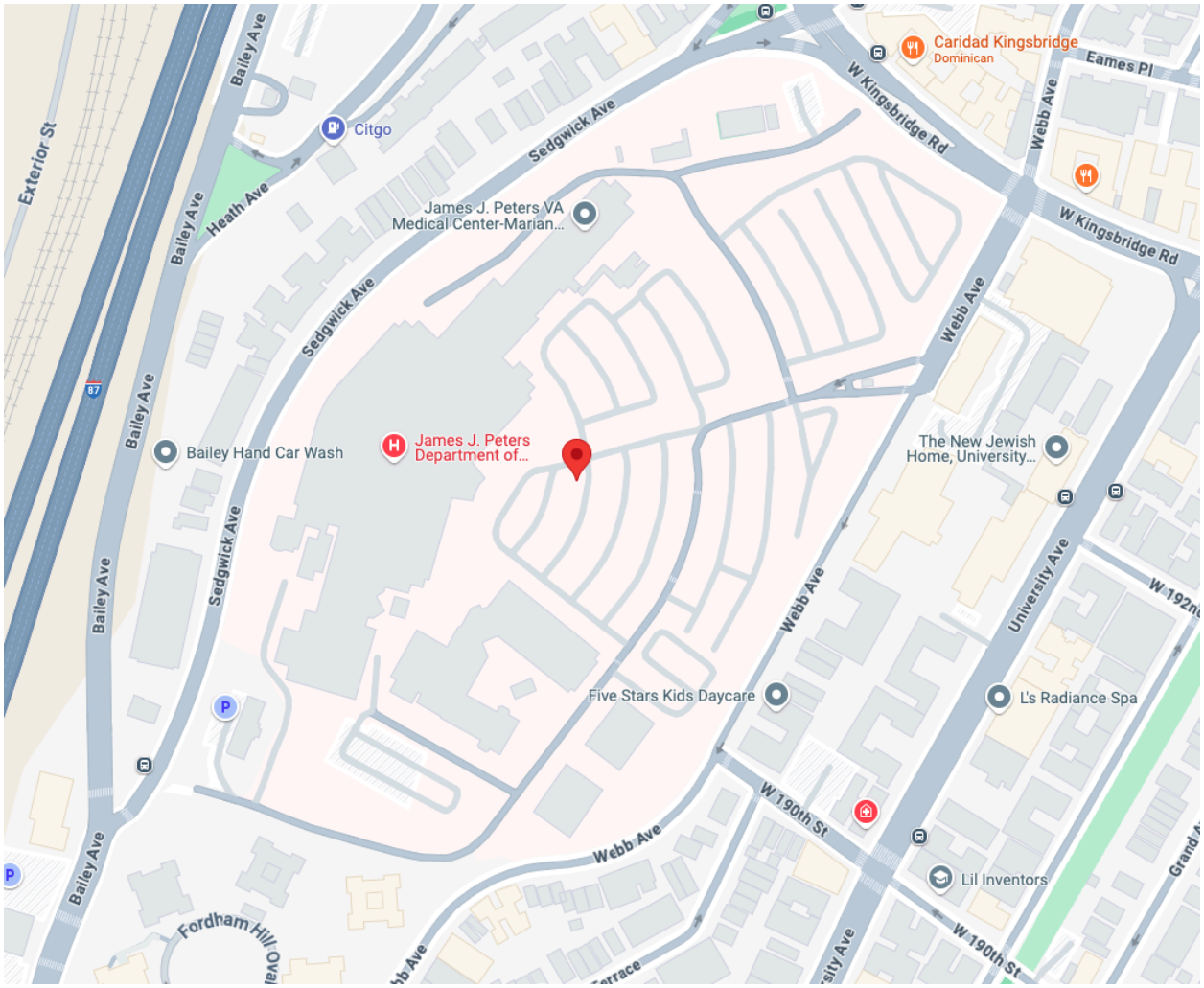
Impact of Misclassification:

- **Ratios Affected**:
  - High r9_taxclass (373.64) and r3_taxclass (186.80) suggest the property's valuation deviates significantly from benchmarks in TAXCLASS 1.
  - These anomalies may be exacerbated by the misclassification of the roof space as a third story.
- **Inconsistencies in Valuation**:
  - The inflated stories count could lead to overvaluation, affecting property tax assessments and market value comparisons.

Recommended Actions:

1. **Inspection of Property Records**:
   - Verify the actual number of stories and reevaluate the classification of attic or roof space.
   - Cross-check zoning and building permits to confirm the structure's specifications.
2. **Reassess Valuation**:
   - Adjust valuation metrics (FULLVAL, AVLAND, and AVTOT) based on the corrected number of stories.
   - Compare the revised data with similar properties in ZIP 11356.
3. **Review Ratios**:
   - Recalculate r3_taxclass and r9_taxclass using accurate data to ensure consistency with comparable properties.

Case Study 2: **Property 170125 (BRONX V.A. Medical Center)**



Property Details:

- **Record ID**: 170125
- **Owner**: BRONX V.A. Medical Center
- **Building Class**: I1 (Hospitals and Health Facilities)
- **Tax Class**: 4 (Commercial/Utility Properties)
- **Lot Dimensions**:
    - **Frontage**: 587 ft

    - **Depth**: 994 ft
- **Stories**: 2
- **Valuation**:
    - **FULLVAL**: $468,100,000
    - **AVLAND**: $9,945,000
    - **AVTOT**: $210,645,000

Observations from the Map:

1. **Property Type**:
    - The property appears to be part of the **James J. Peters VA Medical Center**, a significant hospital and medical facility in the Bronx.
    - This aligns with the **Building Class (I1)** designation.
2. **Surroundings**:
    - The property is part of a larger institutional complex with healthcare facilities, consistent with the classification.

Discrepancies and Anomalies:

1. **Unusual Valuation**:
    - **FULLVAL ($468M)**: While hospitals can have high valuations, the AVLAND ($9.94M) and AVTOT ($210.6M) seem disproportionately low given the size and type of the property. This raises questions about whether the land and improvement assessments are undervalued.
    - **Ratio Analysis**:
        - r2_taxclass (161) and r3_taxclass (462) are abnormally high, indicating valuation discrepancies compared to other properties in TAXCLASS 4.
2. **Stories Count**:
    - The property is reported to have **2 stories**, which is consistent with the facility type, but the footprint (Lot Frontage: 587 ft, Lot Depth: 994 ft) suggests a massive land area that may not fully align with its valuation metrics.
3. **Size Discrepancy**:

- The large lot size and high frontage imply significant land value, but the AVLAND ($9.94M) is low compared to properties of similar size and use in this ZIP code.

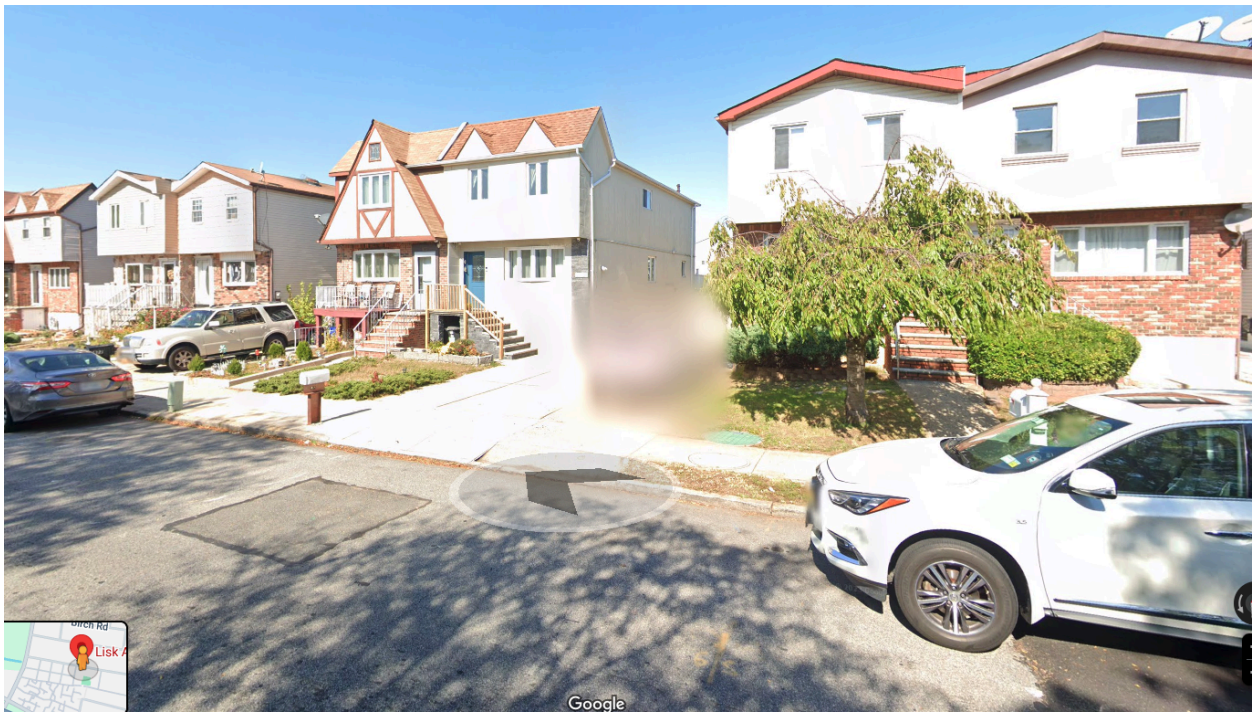
Why This Property is Unusual:

1. **Valuation Mismatch**:
   - The reported valuation metrics suggest underreporting or outdated assessments for land and improvements.
2. **High Ratios**:
   - Ratios r2 and r3 indicate the property valuation deviates significantly from other similar properties in TAXCLASS 4.

Recommended Actions:

1. **Review Valuation Records**:
   - Investigate the methodology used to calculate AVLAND and AVTOT.
   - Determine whether these figures are updated and reflective of the property's current use and market value.
2. **Cross-Check with Comparable Properties**:
   - Compare with other institutional facilities in ZIP 10468 and TAXCLASS 4 to identify patterns of under- or overvaluation.
3. **Inspection**:
   - Conduct a physical inspection to verify the structure and its use align with reported data.

Case Study 3: **Property 979038 (Located on Lisk Avenue)**



Property Details:

- **Record ID**: 979038
- **Owner**: CITY WEST H.O.A.
- **Building Class**: Z0 (Miscellaneous)
- **Tax Class**: 1 (Residential)
- **Lot Dimensions**:
    - **Frontage**: 130 ft
    - **Depth**: 201 ft
- **Stories**: 1
- **Valuation**:
    - **FULLVAL**: $3,670
    - **AVLAND**: $89

- **AVTOT**: $92

Observations from the Map:

1. **Building Type**:
   - The property is part of a residential neighborhood with low-density housing. The homes appear to be typical two-story single-family residences.
   - The property itself, listed as belonging to CITY WEST H.O.A., does not visually match the description of a single-story property.
2. **Surroundings**:
   - The neighborhood consists of well-maintained homes with driveways and landscaped yards. This is consistent with the TAXCLASS of 1 (Residential).

Discrepancies and Anomalies:

1. **Valuation Issues**:
   - **FULLVAL ($3,670)**: This value is **extremely low**, especially for a residential property in a developed neighborhood.
   - **AVLAND ($89)** and **AVTOT ($92)**: These values are unreasonably low and do not reflect the typical land or total valuation for properties in this area.
2. **Lot and Stories**:
   - The reported **1 story** does not align with the visible structure, which appears to be a **two-story building**. This discrepancy could impact the calculated FULLVAL and ratios.
   - The large lot size (Frontage: 130 ft, Depth: 201 ft) should translate to a higher land valuation than $89.
3. **Unusual Ratios**:
   - Ratios r1 to r9 for this property include multiple zeros, indicating possible errors or missing data in the reported values.

Why This Property is Unusual:

1. **Underreporting of Valuation**:

- The extremely low FULLVAL, AVLAND, and AVTOT suggest data inaccuracies or deliberate underreporting.

2. **Classification Errors**:

- The property is misclassified as a single-story structure, which further skews the valuation metrics.

Recommended Actions:

1. **Verify Property Use**:

- Investigate whether this property serves as part of a Homeowners' Association (H.O.A.) or has another specific use not captured in the data.

2. **Reassess Valuation**:

- Update FULLVAL, AVLAND, and AVTOT based on current market rates and property size.

3. **Correct Classification**:

- Review the building's classification to ensure it accurately reflects its structure and use.

Summary of Findings

By using anomaly scores to rank and examine properties, we effectively identify and prioritize unusual records for further investigation. The three case studies demonstrate the utility of combining multiple anomaly detection methods to flag irregularities, providing a robust framework for fraud detection in property valuation data. This process not only highlights discrepancies but also offers actionable insights for remediation and policy adjustments.

Section 8. Summary

Overview of the Steps

This project aimed to detect irregularities and potential fraud within New York City's property tax records by analyzing a dataset of over one million records with 32 features. The analysis followed a structured approach, starting with data cleaning to address missing values and applying exclusions, such as government-owned properties, to narrow the scope. Outliers were intentionally retained as they could indicate fraud rather than errors. Feature engineering introduced variables like size-to-value ratios and standardized metrics to capture nuanced relationships between property characteristics and valuations. Principal Component Analysis (PCA) was applied to reduce dimensionality, retaining the most critical features while improving computational efficiency. Two scoring methods were then developed: Minkowski distance-based scoring to flag deviations from dataset norms and autoencoder reconstruction errors to identify poorly reconstructed records. These scores were normalized, ranked, and combined into a single anomaly detection metric.

Summary of Results and Findings

The results revealed diverse patterns in property records, with final scores showing a multi-modal distribution, reflecting clusters of different property types or valuation tiers. High anomaly scores highlighted properties with irregular relationships, such as underreported lot sizes, implausible building dimensions, or unrealistically low monetary values. Visualizations, including histograms and boxplots, demonstrated that the scoring system was robust, with no extreme outliers skewing the results. Correlation analysis identified key variables, including value-to-size ratios, tax classifications, and geographic patterns, as primary contributors to anomalous scores. Three detailed case studies illustrated the framework's efficacy in identifying suspicious records, such as properties with zero lot dimensions or valuations significantly deviating from their expected range.

Improvements Based on Client Feedback

To improve the system, several enhancements are proposed based on client feedback. Incorporating additional contextual data, such as historical valuations, market trends, and regional benchmarks, can provide a richer basis for scoring. Adjusting thresholds for anomaly scores and experimenting with advanced models, such as hybrid detection systems, could improve the model's sensitivity to nuanced patterns. User-friendly visualizations and interactive dashboards will further aid investigators in exploring flagged records, while iterative feedback loops with domain experts will refine variable selection and exclusion criteria over time. Finally,

optimizing the computational framework for scalability will ensure that the system remains adaptable to larger datasets or real-time applications.

Conclusion

By addressing these areas, the fraud detection framework can better support the City of New York in identifying and mitigating property tax fraud. These improvements will enhance precision, usability, and scalability, ensuring a fairer and more equitable tax system.

Section 9. Appendix: DQR

Data Quality Report

1. Data Description

The dataset consists of approximately 1,070,994 property records from New York City, each representing a single property. The data includes both numerical and categorical fields, capturing various property attributes such as dimensions, assessed values, ownership, location, and other relevant characteristics. This dataset is critical for analyzing anomalies in property records that may indicate tax fraud. Below is a summary of key attributes:

2. Summary Table

Numeric Table

Field Name	# records with values	% Populated	# Zeros	Min	Max	Mean	Std	Most common
LTFRONT	1,070,994	100.0%	169,108	0	9.999e+03	3.664e+01	7.403e+01	0
LTDEPTH	1,070,994	100.0%	170,128	0	9.999e+03	8.886e+01	7.640e+01	100
STORIES	1,014,730	94.7%	0	1	1.190e+02	5.010e+00	8.370e+00	2
FULLVAL	1,070,994	100.0%	13,007	0	6.150e+09	8.743e+05	1.158e+07	0
AVLAND	1,070,994	100.0%	13,009	0	2.668e+09	8.507e+04	4.057e+06	0
AVTOT	1,070,994	100.0%	13,007	0	4.668e+09	2.272e+05	6.878e+06	0
EXLAND	1,070,994	100.0%	491,699	0	2.668e+09	3.642e+04	3.982e+06	0
EXTOT	1,070,994	100.0%	432,572	0	4.668e+09	9.119e+04	6.508e+06	0
BLDFRONT	1,070,994	100.0%	228,815	0	7.575e+03	2.304e+01	3.558e+01	0
BLDDEPTH	1,070,994	100.0%	228,853	0	9.393e+03	3.992e+01	4.271e+01	0
AVLAND2	282,726	26.4%	0	3	2.371e+09	2.462e+05	6.179e+06	2,408
AVTOT2	282,732	26.4%	0	3	4.501e+09	7.139e+05	1.165e+07	750
EXLAND2	87,449	8.2%	0	1	2.371e+09	3.512e+05	1.080e+07	2,090
EXTOT2	130,828	12.2%	0	7	4.501e+09	6.568e+05	1.607e+07	2,090

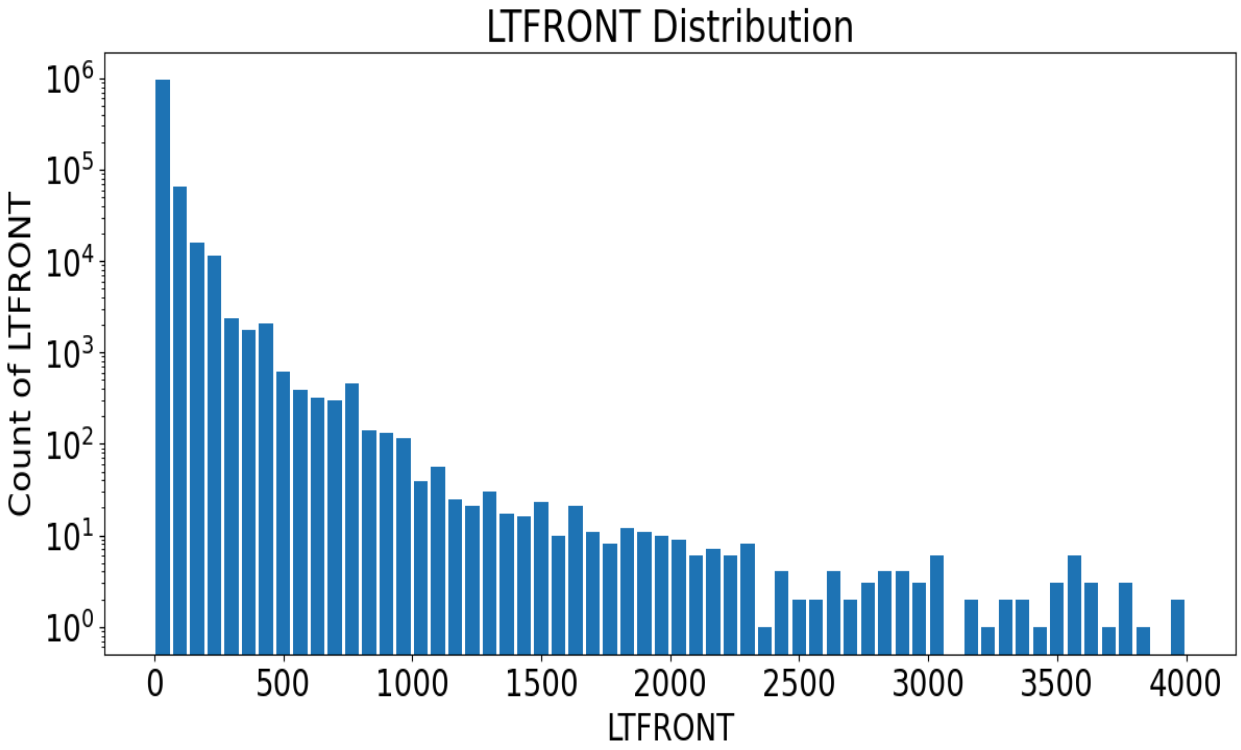
Categorical Table

Field Name	# records with values	% Populated	# Zeros	# Unique Values	Most Common
RECORD	1,070,994	100.0%	0	1,070,994	1
BBLE	1,070,994	100.0%	0	1,070,994	1000010101
BORO	1,070,994	100.0%	0	5	4
BLOCK	1,070,994	100.0%	0	13,984	3944
LOT	1,070,994	100.0%	0	6,366	1
EASEMENT	4,636	0.4%	0	12	E
OWNER	1,039,249	97.0%	0	863,347	PARKCHESTER PRESERVAT
BLDGCL	1,070,994	100.0%	0	200	R4
TAXCLASS	1,070,994	100.0%	0	11	1
EXT	354,305	33.1%	0	3	G
EXCD1	638,488	59.6%	0	129	1017
STADDR	1,070,318	99.9%	0	839,280	501 SURF AVENUE
ZIP	1,041,104	97.2%	0	196	10314
EXMPTCL	15,579	1.5%	0	14	X1
EXCD2	92,948	8.7%	0	60	1017
PERIOD	1,070,994	100.0%	0	1	FINAL
YEAR	1,070,994	100.0%	0	1	2010/11
VALTYPE	1,070,994	100.0%	0	1	AC-TR

3. Visualization of Each Field

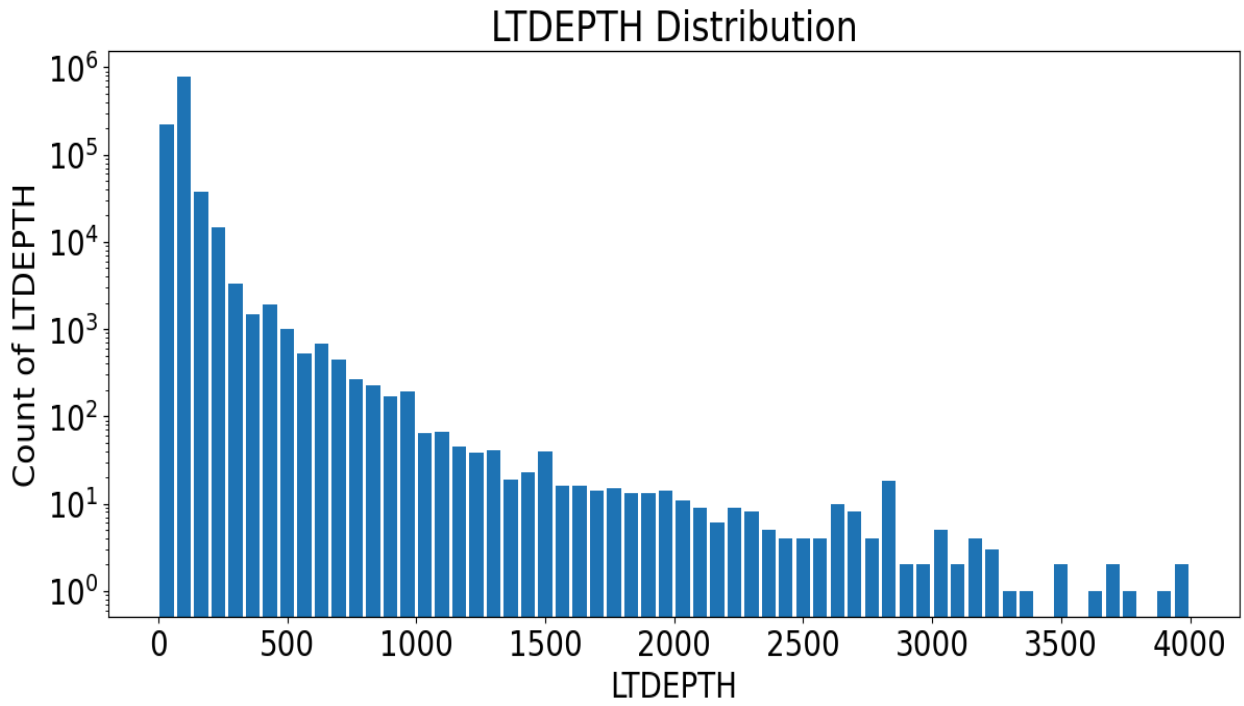
Numeric Fields:

(1) Field Name: LTFRONT



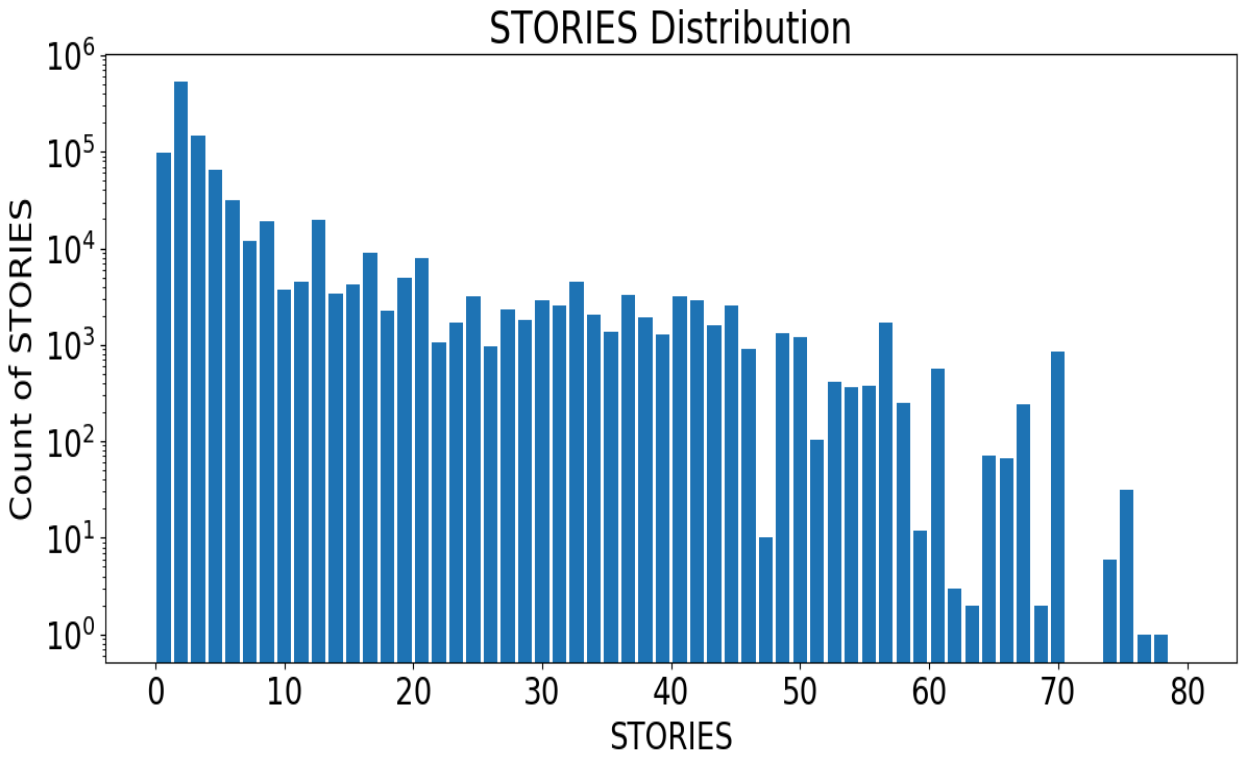
The LTFRONT field represents lot frontage length in feet. The distribution is skewed, with most values near 0 and a few exceeding 2,000 feet. The logarithmic scale highlights the sharp decline in frequency as LTFRONT increases, indicating most properties have small frontage lengths.

(2) Field Name: LTDEPTH



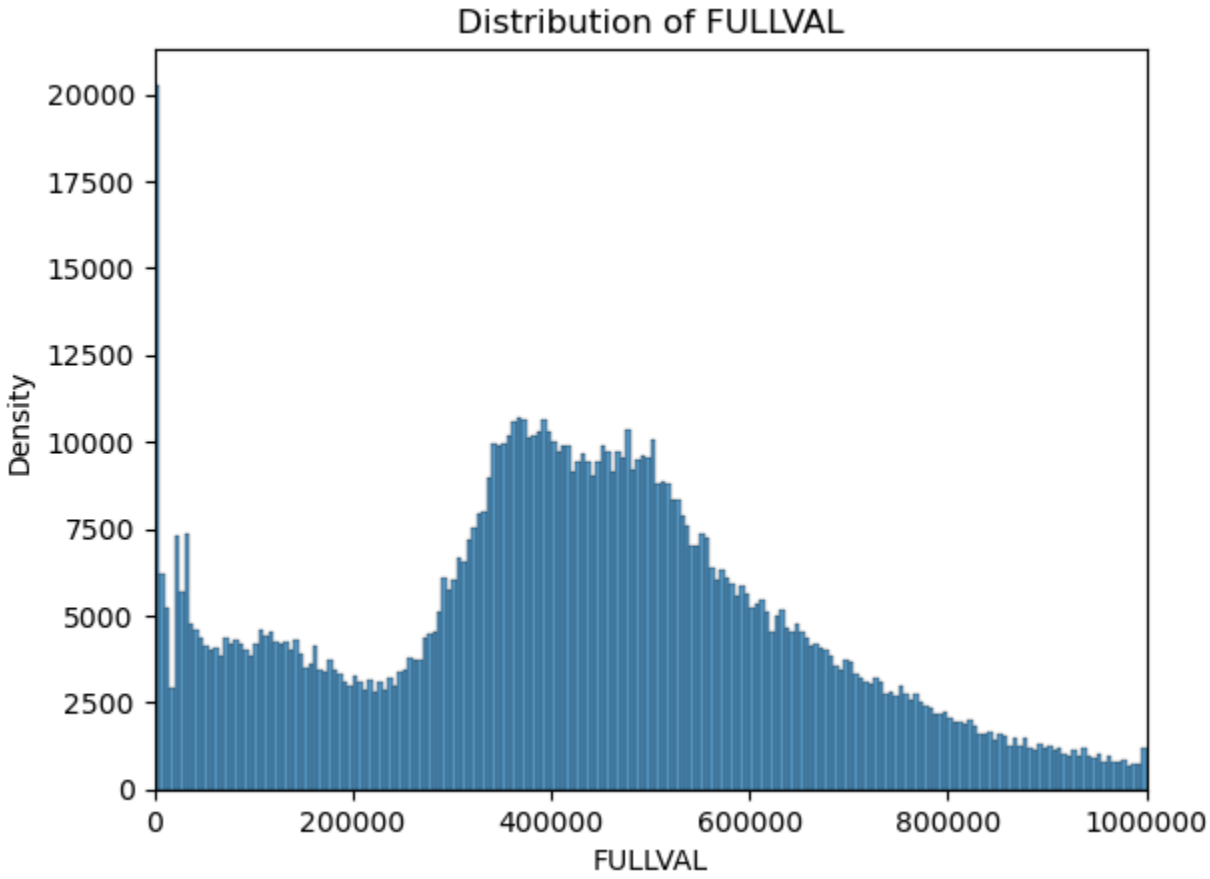
The LTDEPTH field represents lot depth in feet. The distribution is skewed, with the majority of values clustered near 0 and a gradual decline as depth increases. The logarithmic scale emphasizes the prevalence of smaller lot depths, with fewer lots exceeding 2,000 feet.

(3) Field Name: STORIES



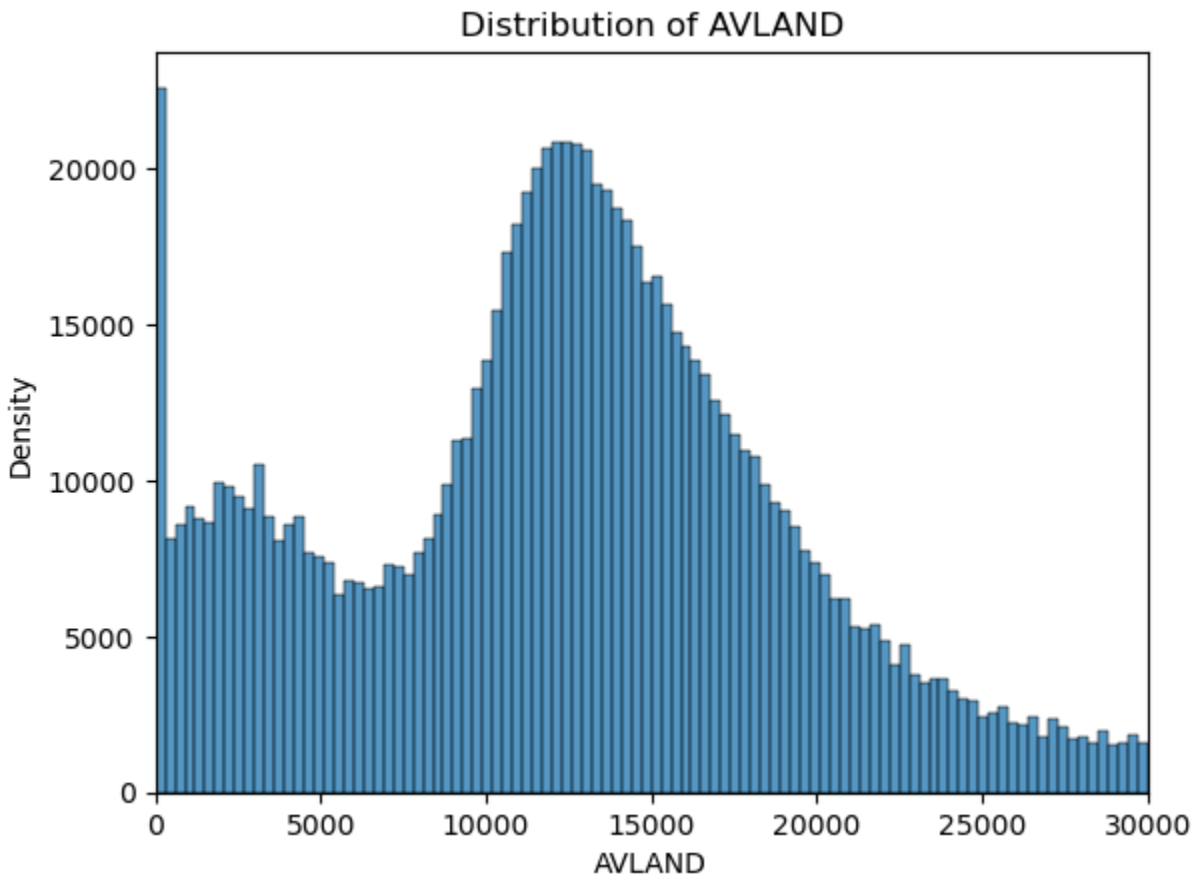
The STORIES field represents the number of floors in a building. The distribution is right-skewed, with most buildings having fewer than 10 stories. A small number of properties have more than 50 stories, with the tallest buildings reaching 80 stories. The logarithmic scale highlights the dominance of low-rise structures in the dataset.

(4) Field Name: FULLVAL



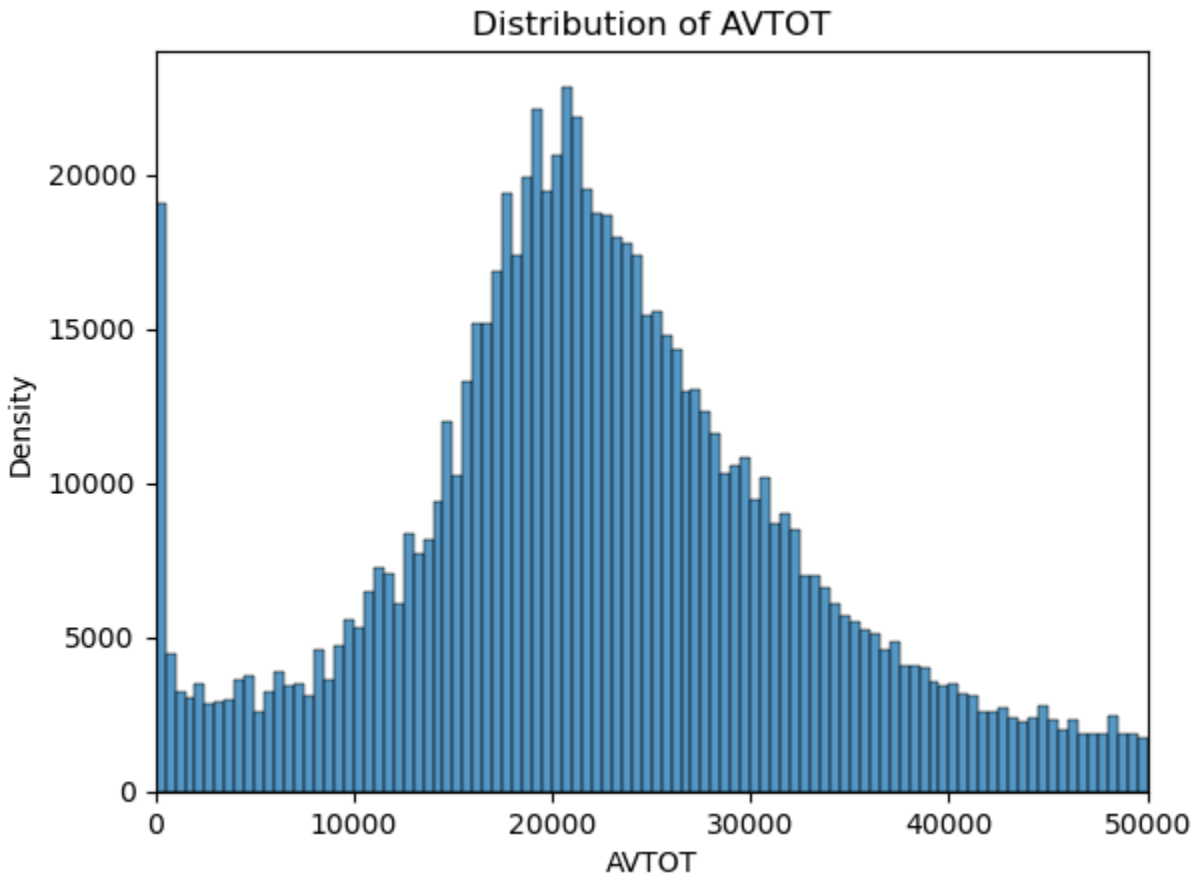
The distribution of the FULLVAL field reveals a bimodal pattern, with a significant number of small values concentrated near zero and a second peak around 400,000. The density gradually declines as values approach the upper limit of 1,000,000. This distribution highlights variability in property values, with most properties falling below 500,000.

(5) Field Name: AVLAND



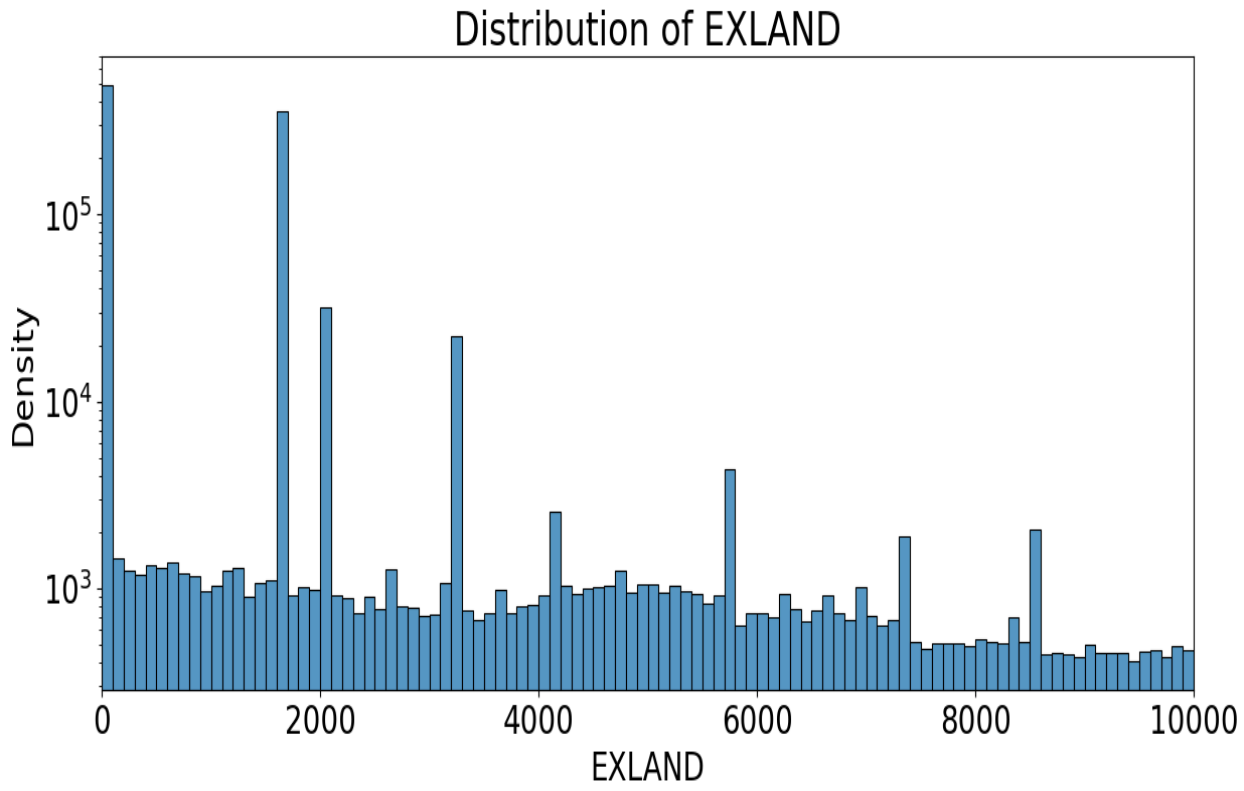
The distribution of AVLND exhibits a clear peak around 10,000, indicating the most common land values in the dataset. The density decreases gradually beyond this point, with fewer occurrences approaching the upper limit of 30,000. The data highlights a concentration of smaller values, with a visible skewness toward lower land values.

(6) Field Name: AVTOT



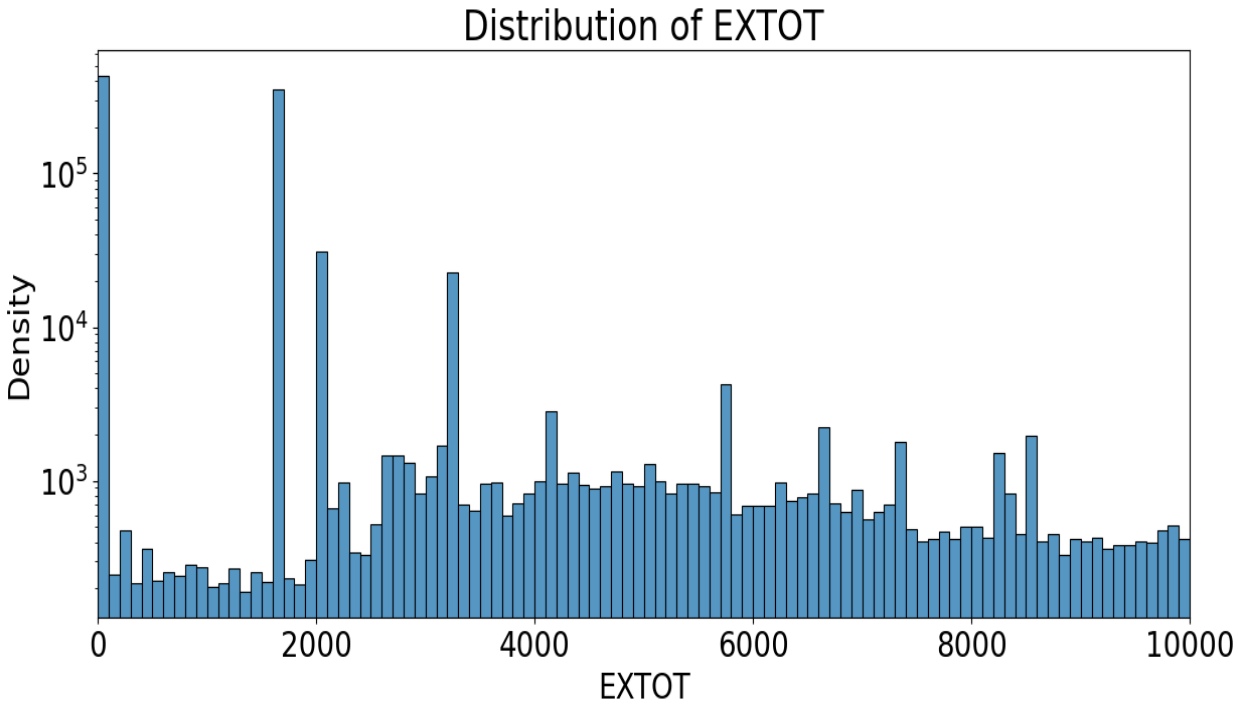
The updated AVTOT plot showcases a clear unimodal distribution with the peak centered around 20,000 and a well-distributed range extending up to 50,000. This adjusted x-axis effectively focuses on the bulk of the data, excluding outliers, and enhances the interpretability of the density curve by highlighting the most relevant data points.

(7) Field Name: EXLAND



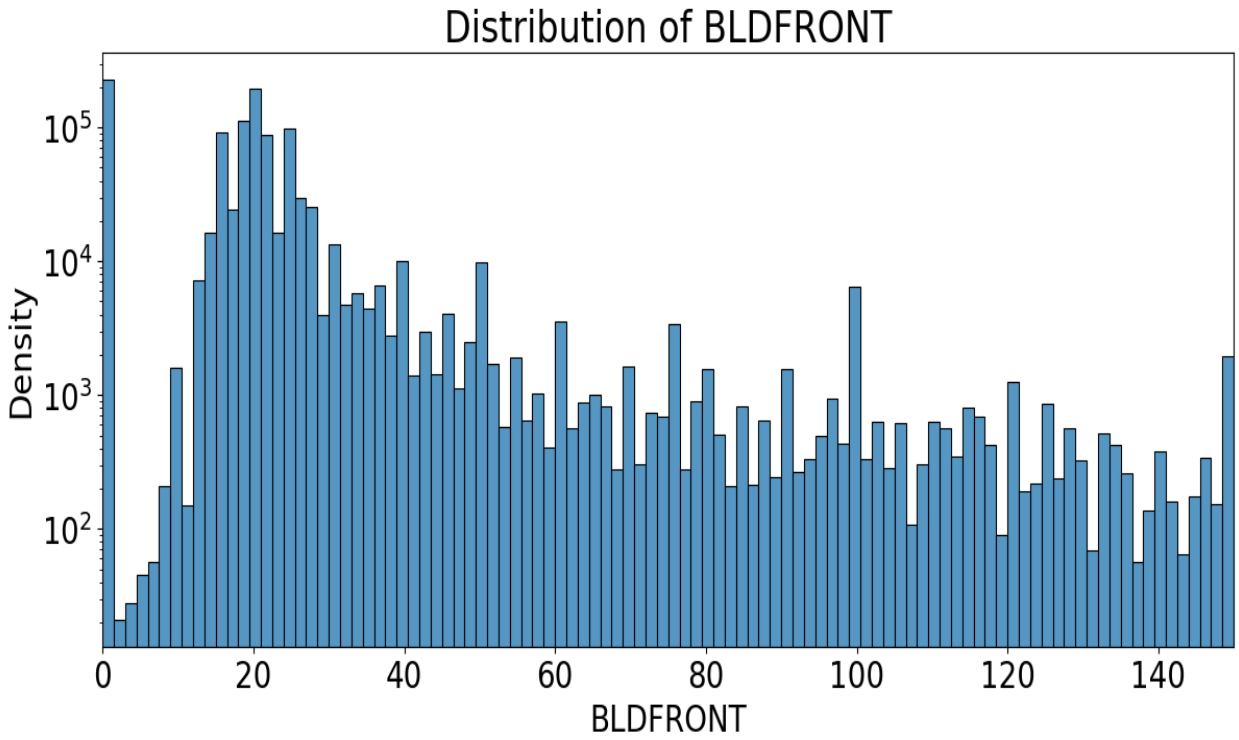
The EXLAND field represents land-related values with a multimodal distribution, displaying distinct spikes at regular intervals. These peaks likely correspond to specific groupings or thresholds in the data. The histogram highlights the concentration of values around these spikes, suggesting patterns or classifications worth further exploration.

(8) Field Name: EXTOT



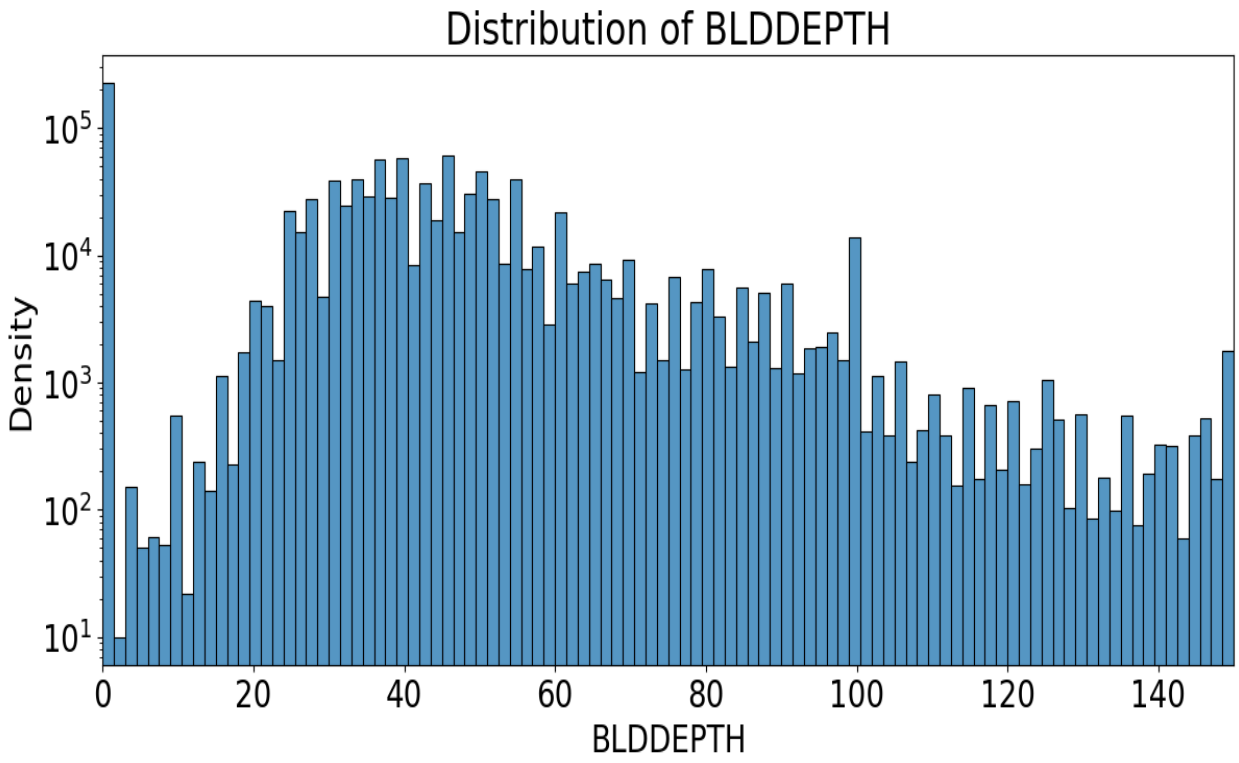
The EXTOT field shows a multimodal distribution with distinct peaks at regular intervals, indicating clusters or specific groupings within the data. The histogram highlights significant variation in density, with notable spikes around certain values, suggesting potential thresholds or classifications influencing the data distribution.

(9) Field Name: BLDFRONT



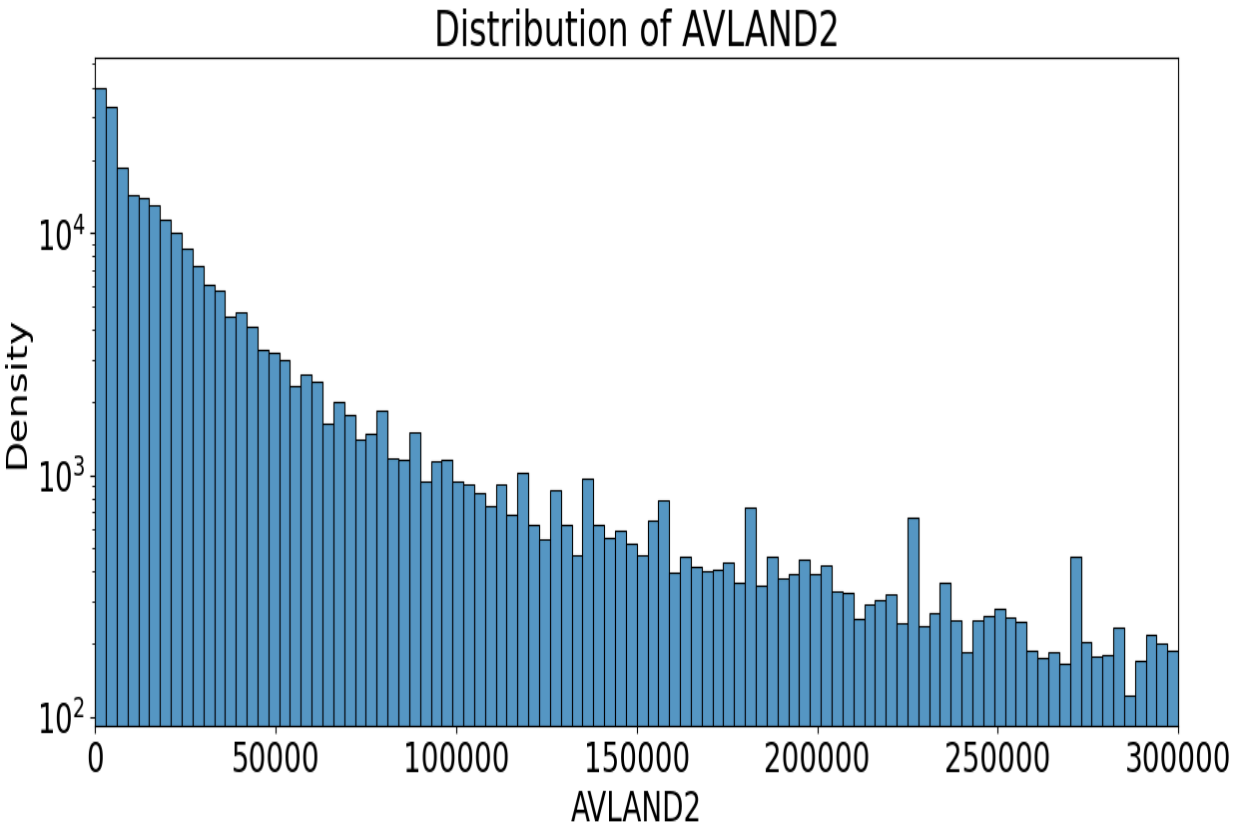
The BLDFRONT field represents building frontage dimensions and exhibits a right-skewed distribution, with a majority of values clustered below 50. The histogram also reveals several peaks at regular intervals, suggesting possible standard frontage sizes.

(10) Field Name: BLDDEPTH



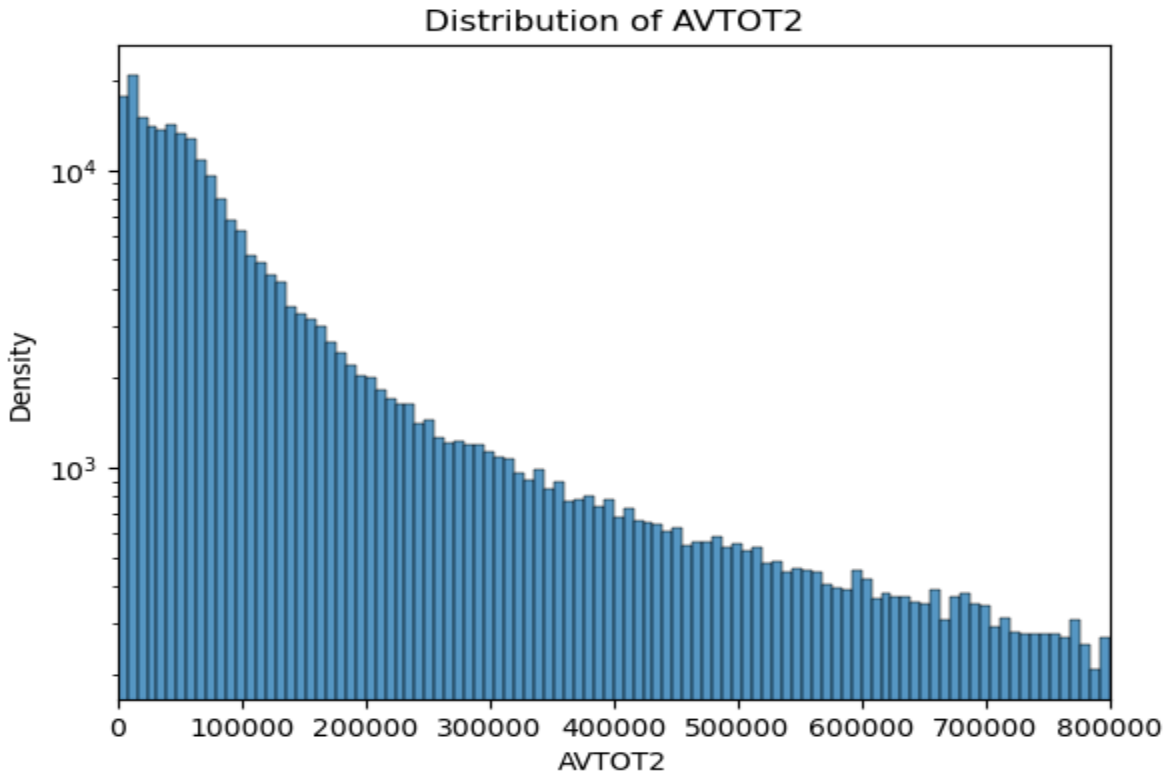
The BLDDEPTH field represents the depth of buildings, showing a relatively uniform distribution between 20 and 100, with some peaks indicating common standard depths. The density declines slightly beyond 100, suggesting fewer deeper buildings.

(11) Field Name: AVLAND2



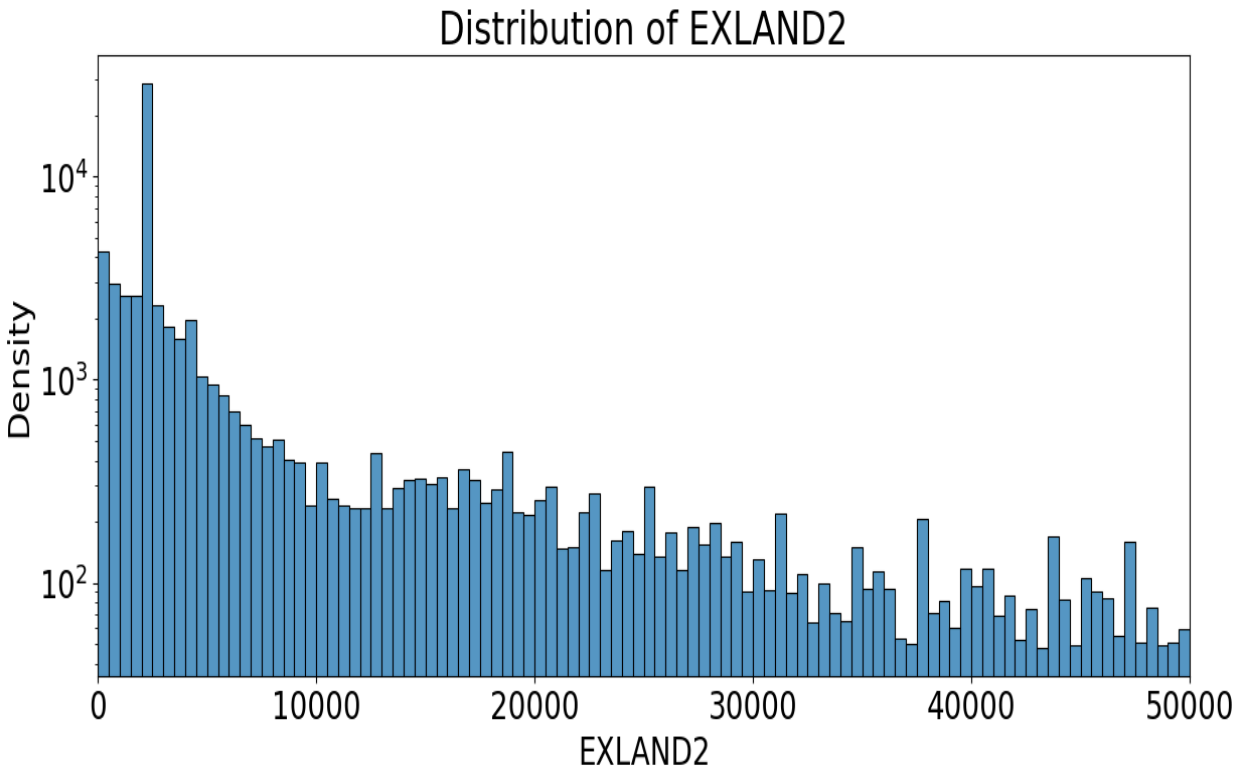
The AVLAND2 field represents a secondary measure of land value. Its distribution is right-skewed, with the majority of values concentrated below 50,000 and a gradual decline as values increase, suggesting a small number of properties with significantly higher land values.

(12) Field Name: AVTOT2



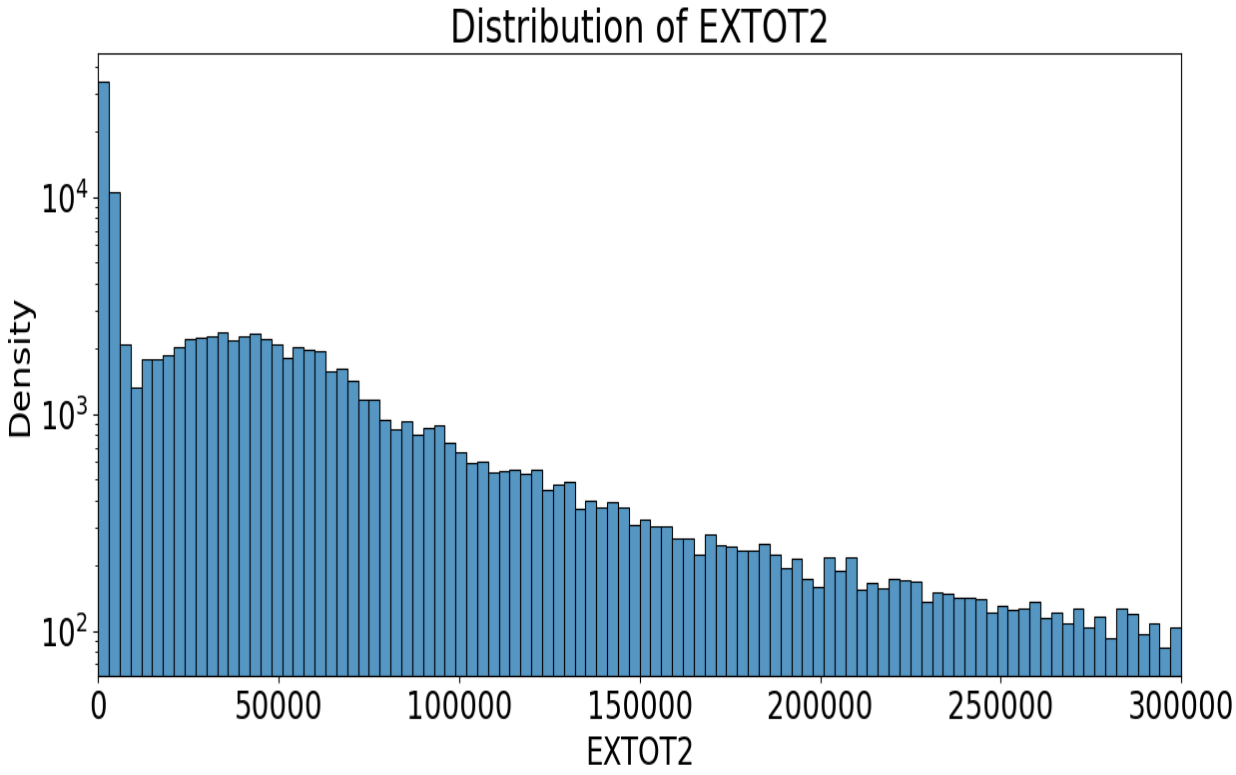
The AVTOT2 field represents the assessed total value of properties. The distribution shows a right-skewed pattern with most values concentrated below 200,000, while a small number of properties have significantly higher values. This highlights the disparity in property assessments within the dataset.

(13) Field Name: EXLAND2



The EXLAND2 field represents the exempt land value of properties. The distribution is highly right-skewed, with most values clustered below 10,000 and a gradual decline for higher values. This indicates that a majority of properties have low exempt land values, while a smaller subset has significantly higher exemptions.

(14) Field Name: EXTOT2



The EXTOT2 field represents the total exemption value of properties. The distribution is right-skewed, with most values concentrated below 50,000 and a gradual tapering off for higher values. This suggests that the majority of properties have relatively low total exemption values, with fewer properties having higher exemptions.

Categorical Field:

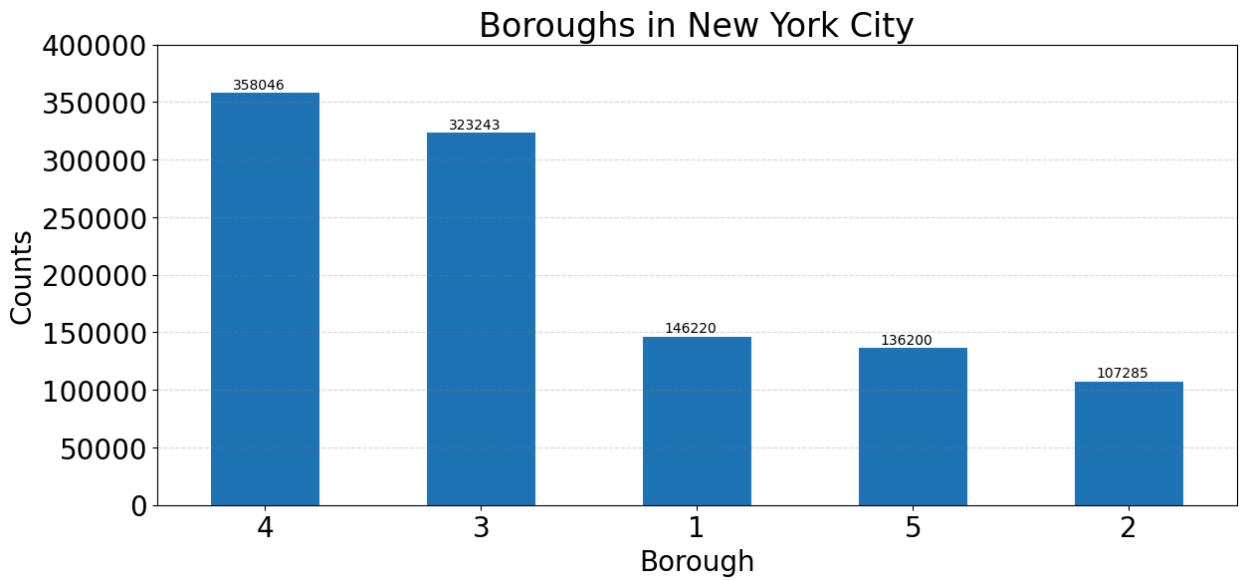
(1) Field Name: RECORD

The RECORD field serves as a unique identifier for each property record in the dataset. It contains 1,070,994 distinct values, with no duplicates or missing values.

(2) Field Name: BBLE

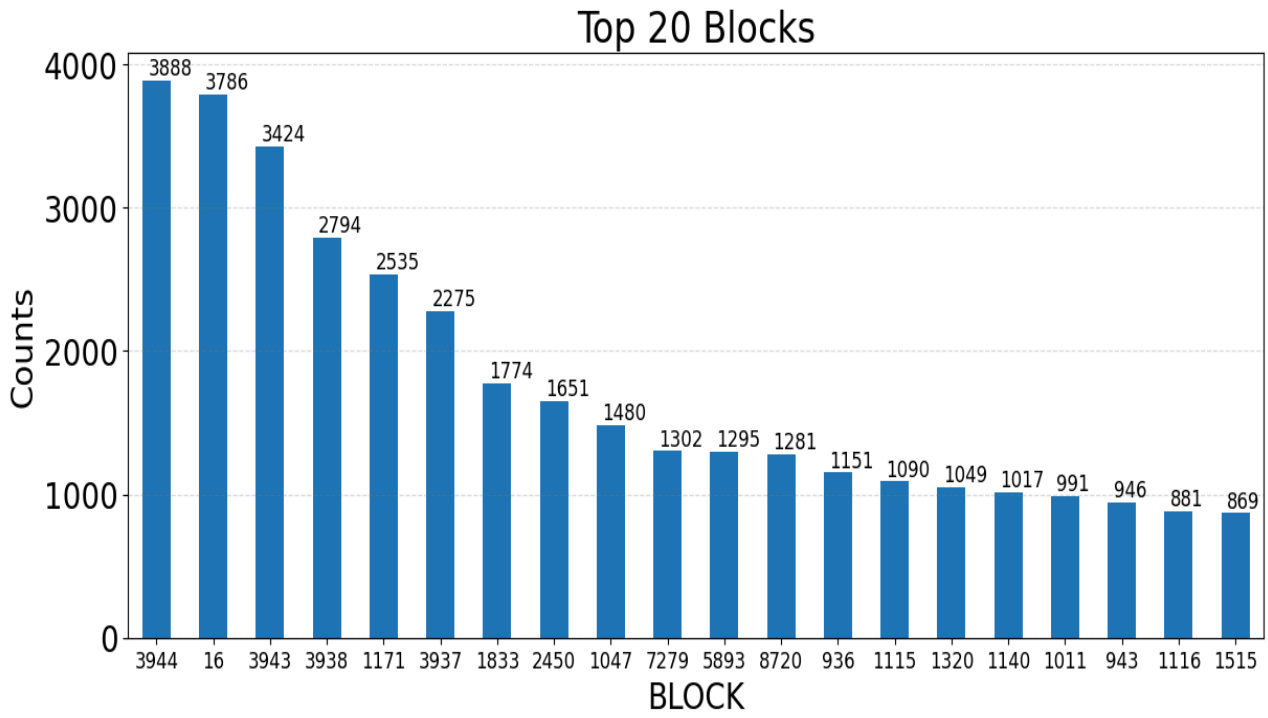
The BBLE field functions as another unique identifier for properties, often used for tax and administrative purposes. Similar to RECORD, it contains 1,070,994 unique entries, with each value corresponding to a specific property in the dataset.

(3) Field Name: BORO



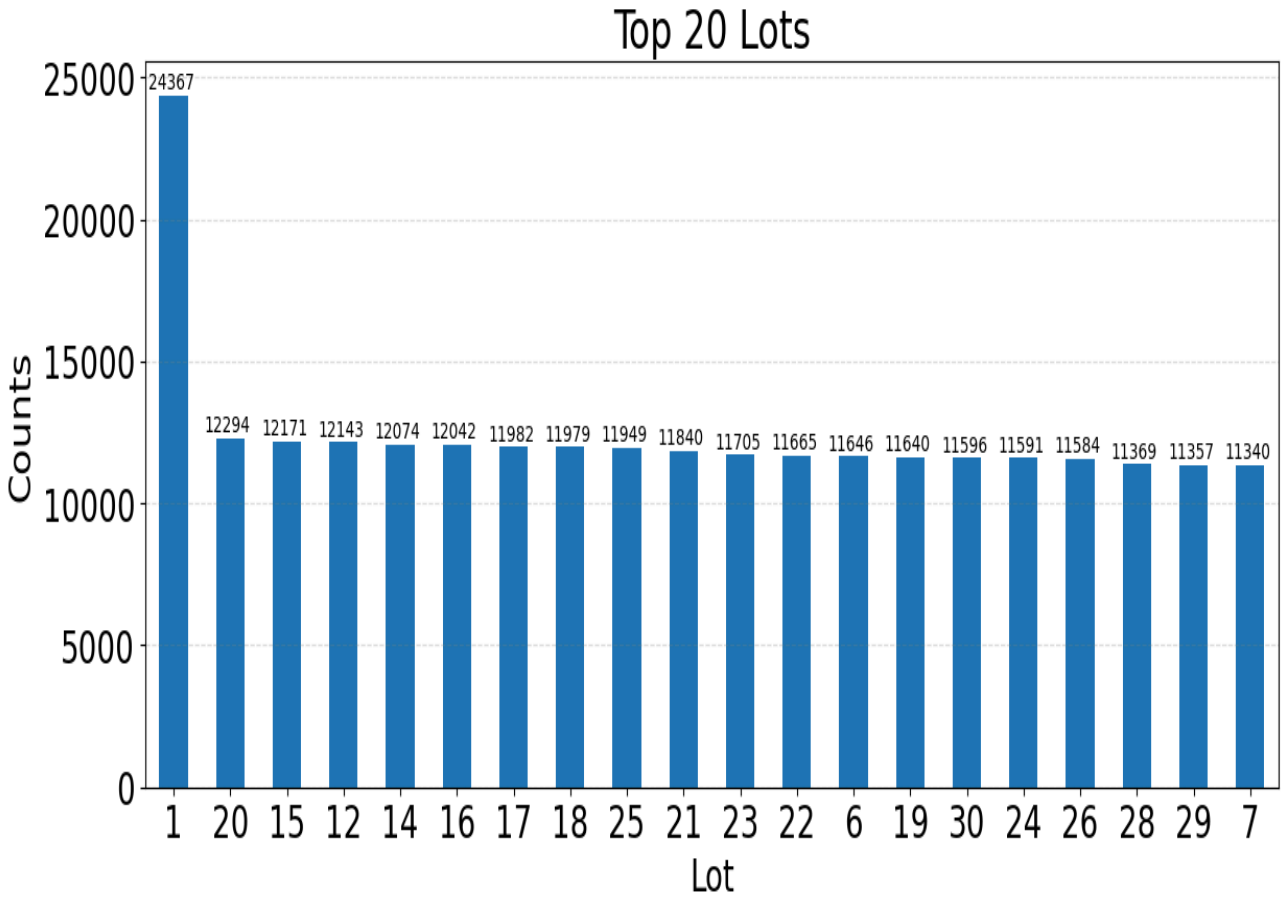
The BORO field represents the five boroughs of New York City: Manhattan (1), Bronx (2), Brooklyn (3), Queens (4), and Staten Island (5). The chart shows that Queens (4) and Brooklyn (3) have the highest property counts, with 350,046 and 321,243 records, respectively. Manhattan (1) and Staten Island (5) have similar counts of 146,220 and 136,260, while the Bronx (2) has the fewest at 107,285. This distribution highlights the geographic diversity of properties in the dataset.

(4) Field Name: BLOCK



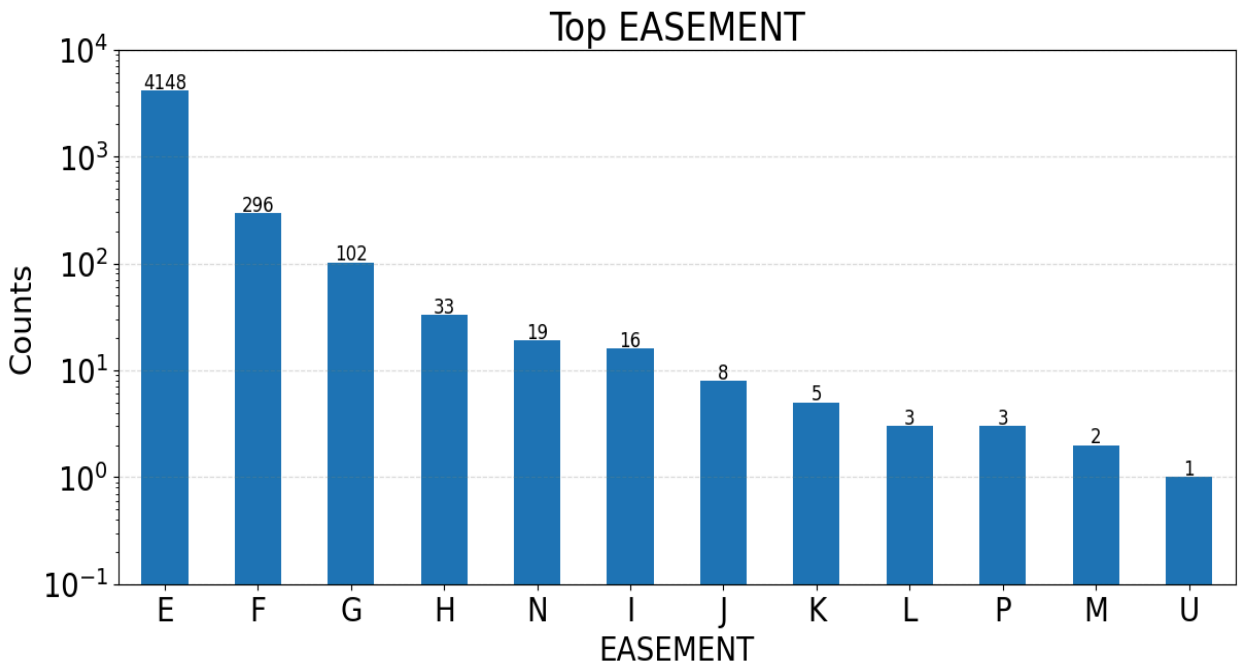
The BLOCK field represents unique block identifiers within New York City boroughs. The chart displays the top 20 blocks by property record count, with block 3944 having the highest count (3,888), followed by blocks 16 and 3786 with 3,786 and 3,424 records, respectively. There is a noticeable drop in record counts after the top three blocks, with blocks like 1116 and 1515 at the lower end, each having fewer than 900 records.

(5) Field Name: LOT



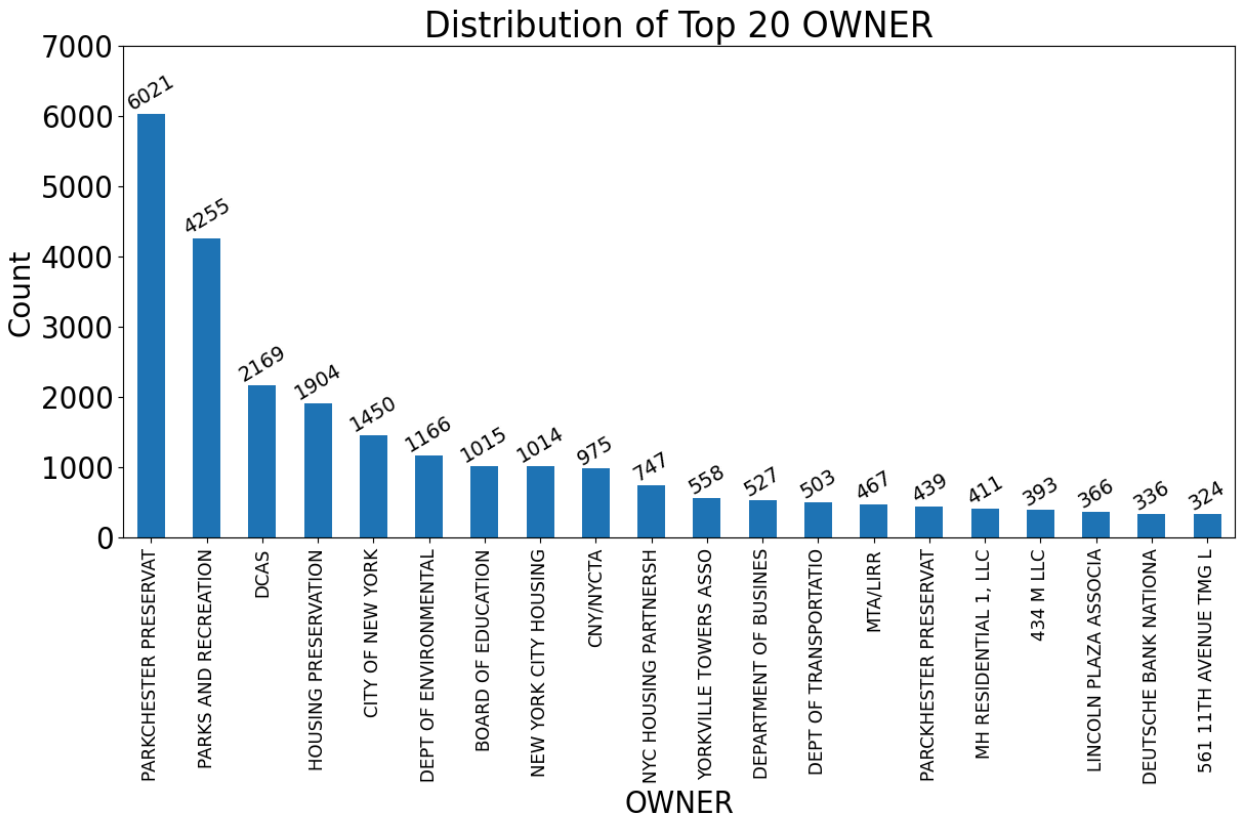
The LOT field represents unique lot identifiers within each block in New York City. The chart shows the top 20 most frequent lots, with lot 1 dominating at 24,367 records, followed by lot 20 with 12,294 and lot 15 with 12,171. The remaining lots show similar counts, ranging from 12,143 to 11,340. This field is critical for analyzing property distribution at a granular level.

(6) Field Name: EASEMENT



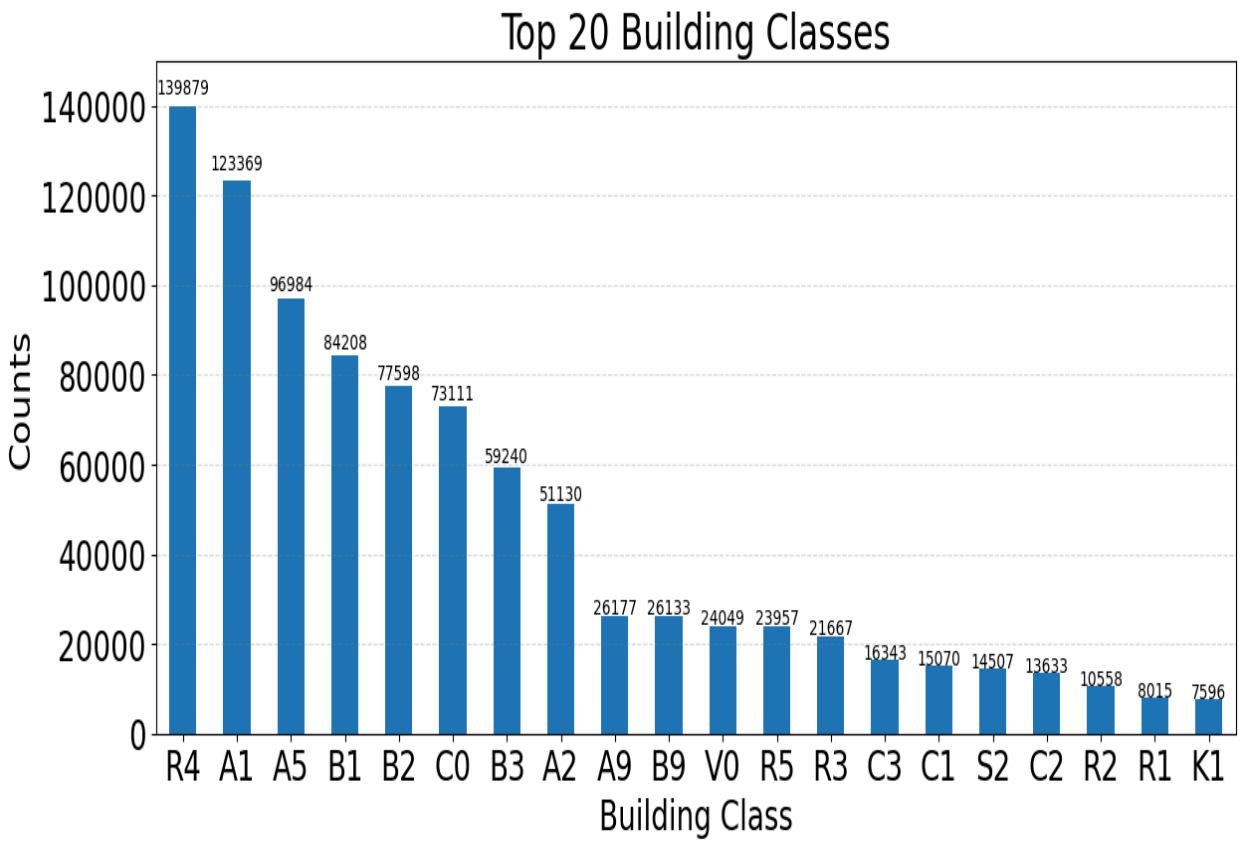
The EASEMENT field represents property easement types. The chart shows the top categories, with E being the most common (4,148 records), followed by F (296 records) and G (102 records). Other categories have significantly fewer records, indicating their rarity in the dataset.

(7) Field Name: OWNER



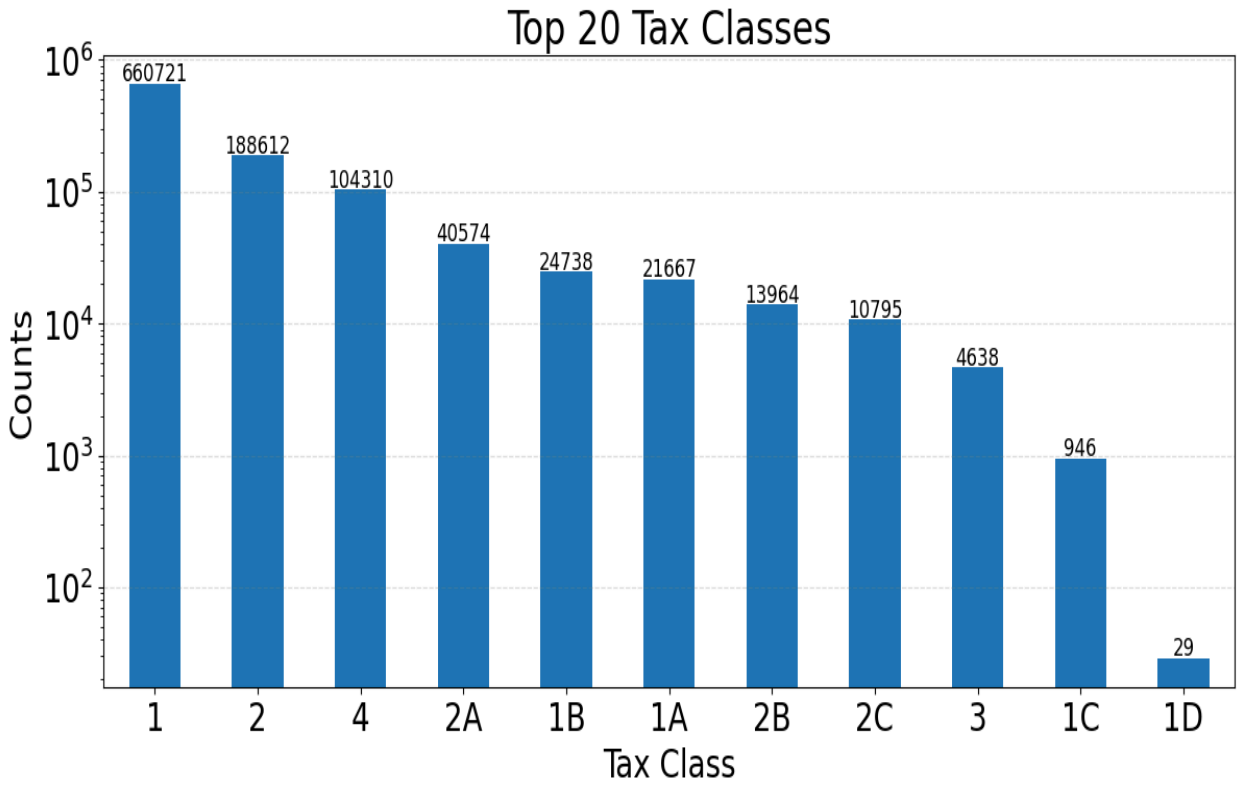
The OWNER field represents property ownership information. The chart highlights the top 20 owners, with “PARKCHESTER PRESERVAT” having the highest count (6,021), followed by “PARKS AND RECREATION” (4,255) and “DCAS” (2,169). Other owners have fewer records, showing a diverse distribution of ownership across the dataset.

(8) Field Name: BLDGCL



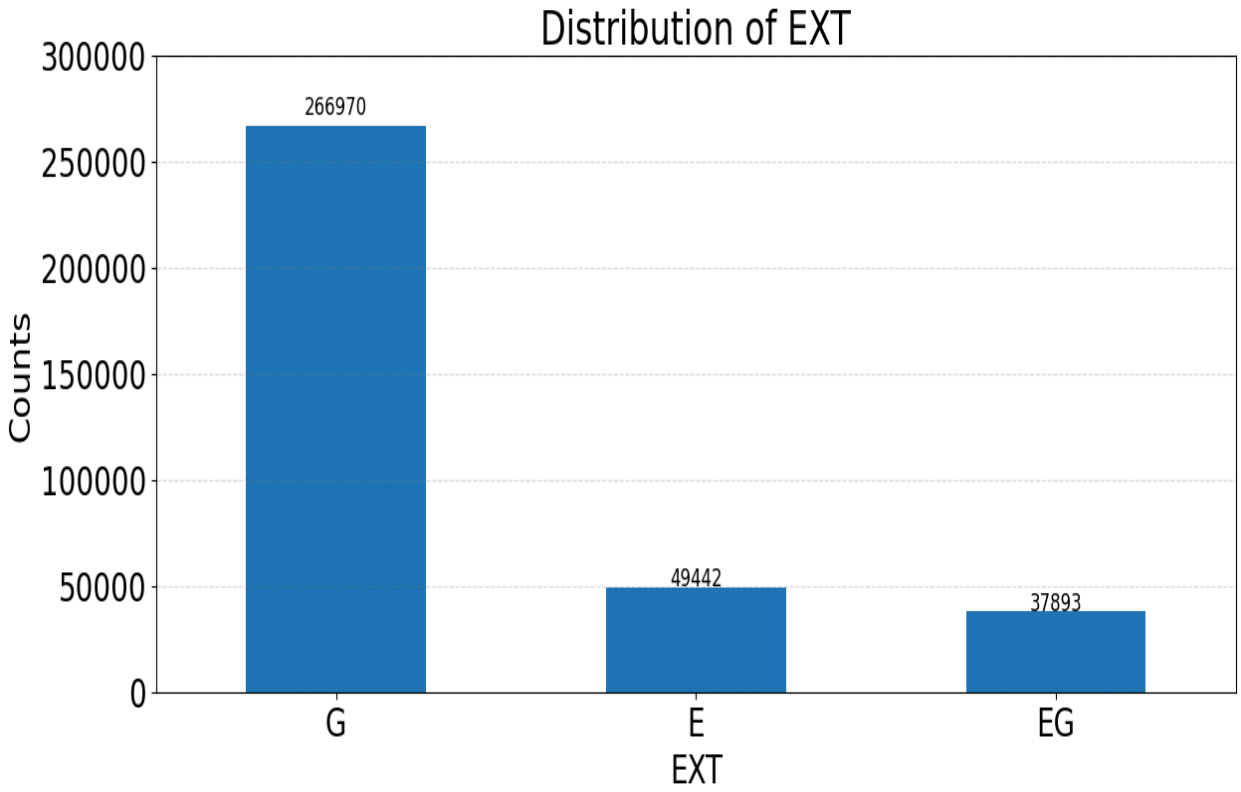
The BLDGCL field represents building class codes, categorizing properties based on their use or type. The chart shows the top 20 building classes, with R4 being the most frequent (139,919 records), followed by A1 (123,369) and A5 (96,084). Other classes, such as B2, C0, and A2, also have significant representation, while the remaining classes exhibit lower counts. This field is critical for understanding property types and their distribution across the dataset.

(9) Field Name: TAXCLASS



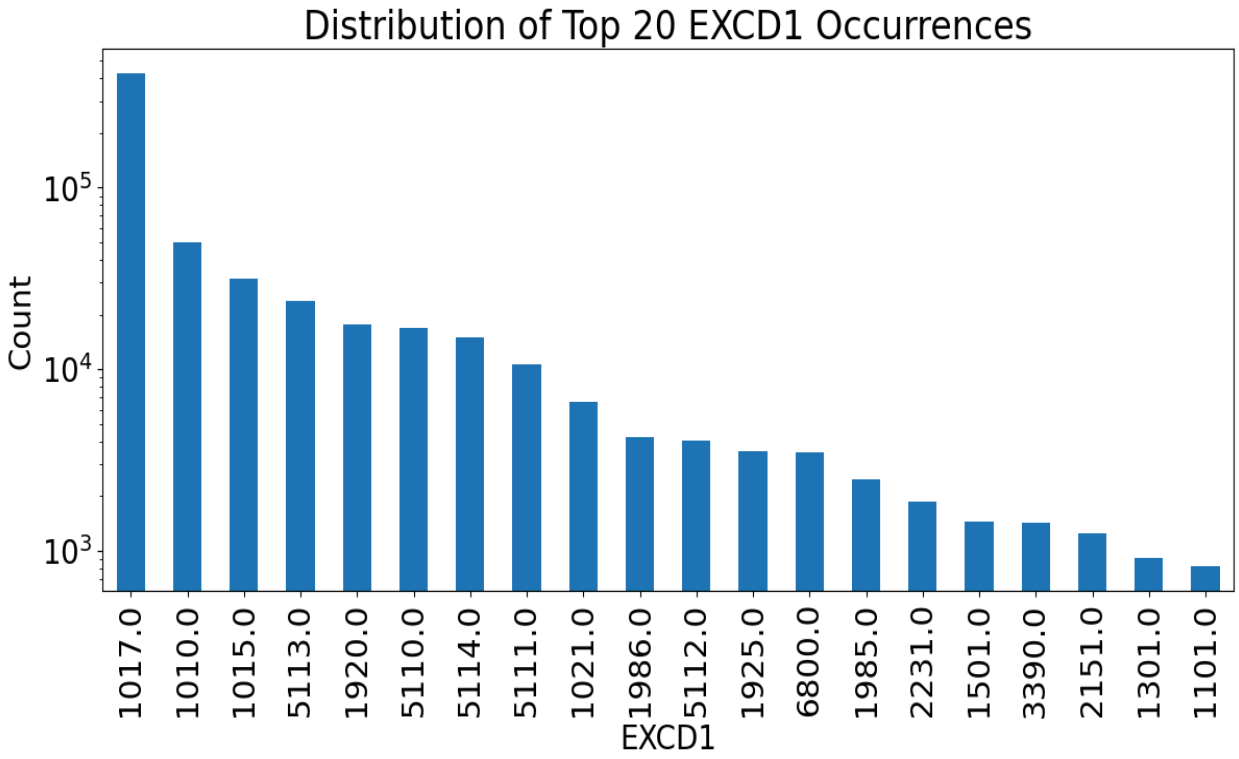
The TAXCLASS field represents the tax classification of properties, which determines tax rates based on property type and use. The chart highlights the top 20 tax classes, with class 1 being the most frequent (660,721 records), followed by classes 2 (188,612) and 4 (104,310). Other tax classes, such as 2A, 1B, and 1A, have fewer records but are still significant. This field is essential for understanding property taxation trends within the dataset.

(10) Field Name: EXT



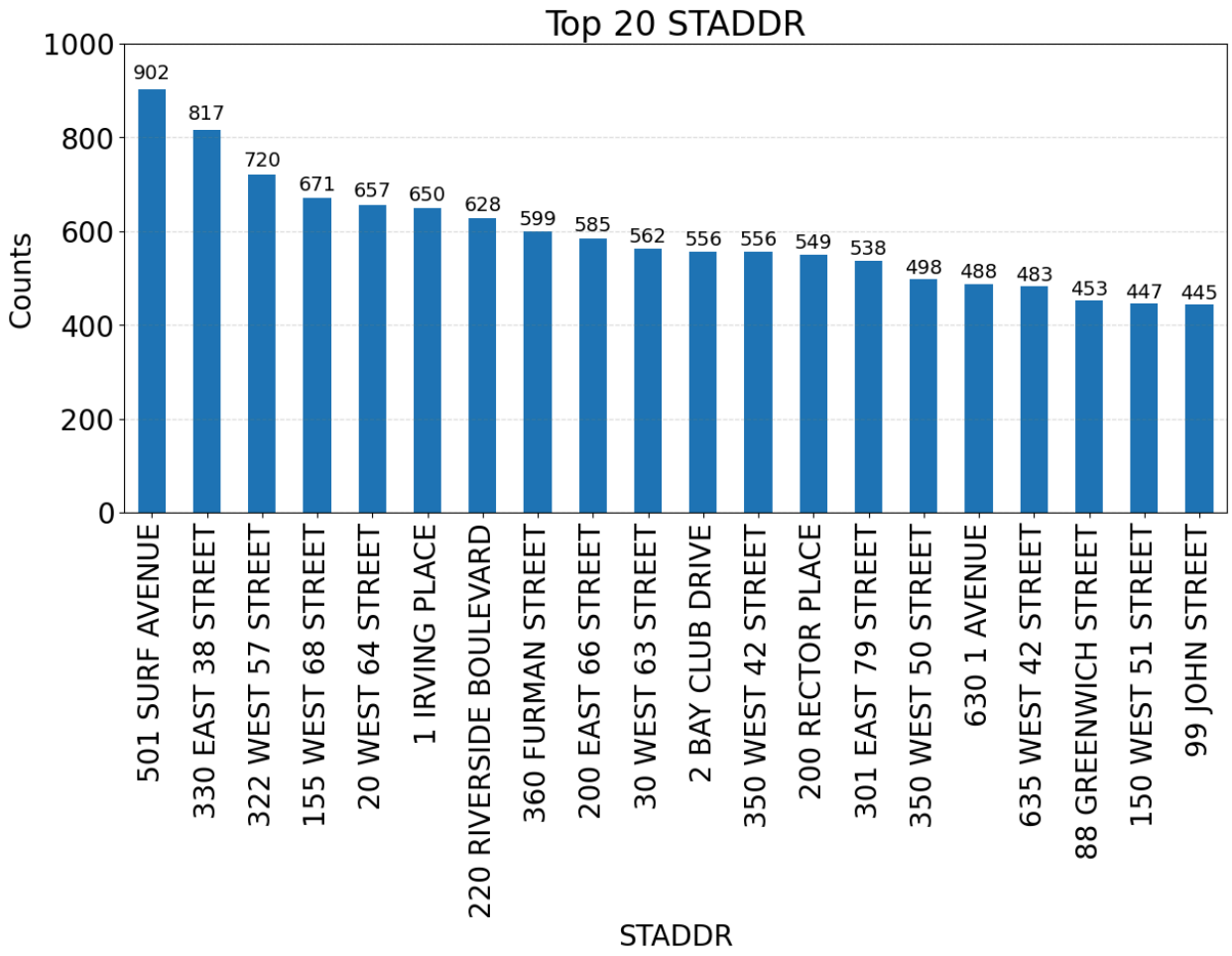
The EXT field represents external property classifications. The chart shows that G is the most common classification with 266,970 records, followed by E with 40,142 records and EG with 37,893 records. This field provides insight into property characteristics based on external features.

(11) Field Name: EXCD1



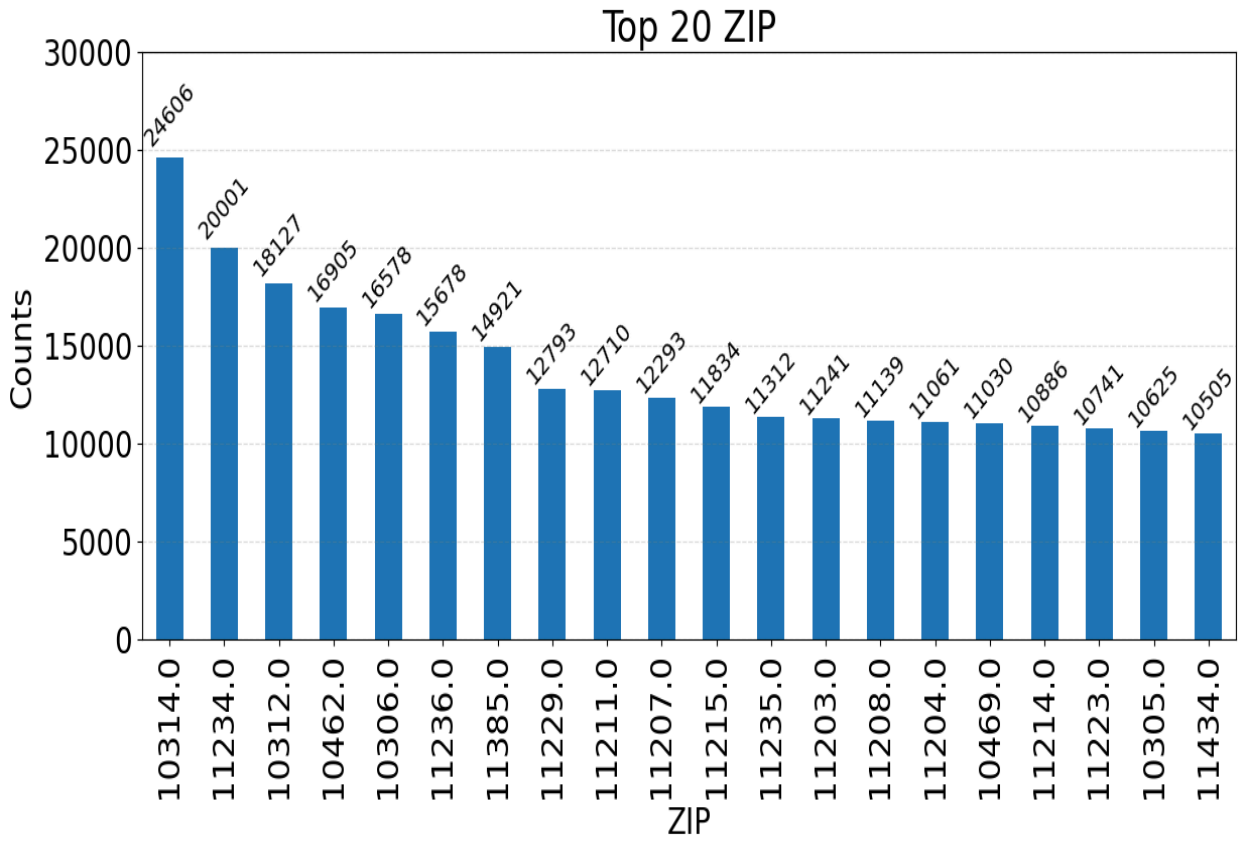
The EXCD1 field represents property exception codes, which provide specific categorizations or annotations. The chart displays the top 20 most frequent codes, with 1017.0 being the most common, followed by 1010.0 and 1015.0. The remaining codes exhibit a gradual decline in frequency, reflecting the diverse range of exception classifications in the dataset.

(12) Field Name: STADDR



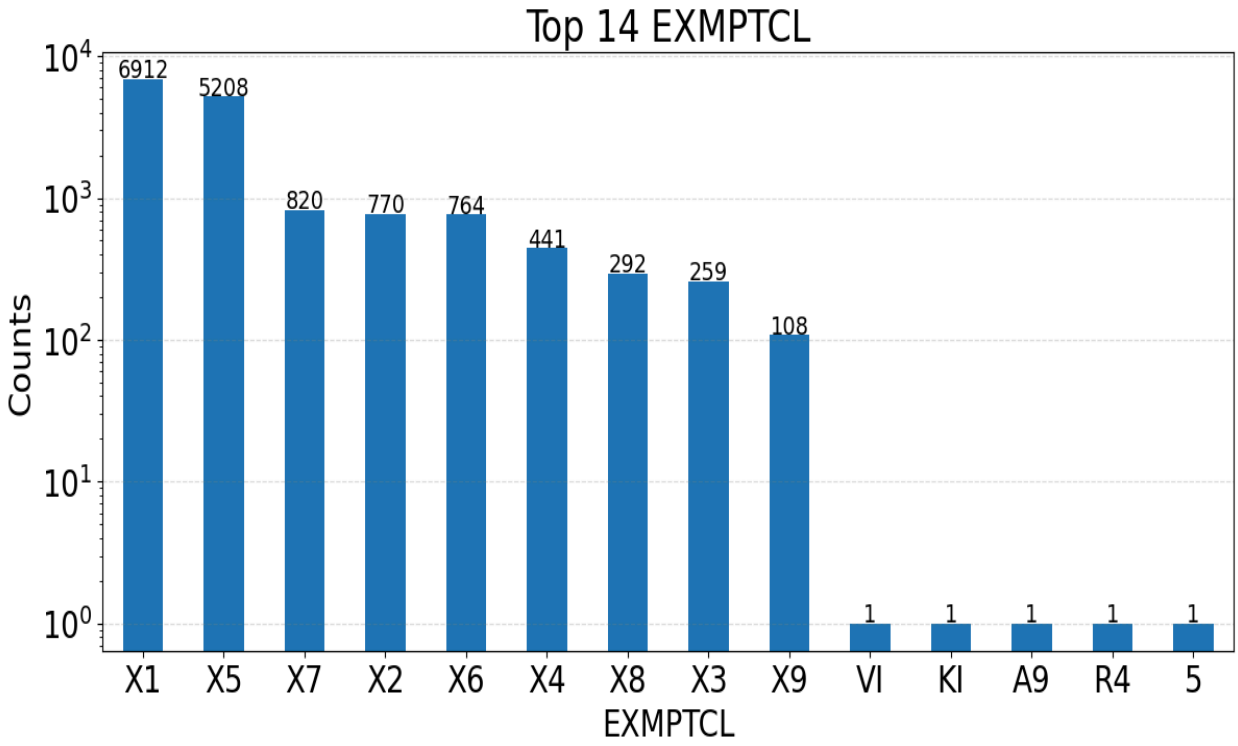
The STADDR field represents street addresses associated with properties. The chart shows the top 20 most frequent addresses, with “501 SURF AVENUE” having the highest count (902 records), followed by “330 EAST 38 STREET” (817 records) and “322 WEST 57 STREET” (720 records). The remaining addresses show a gradual decline in frequency, highlighting the concentration of records at specific locations within the dataset.

(13) Field Name: ZIP



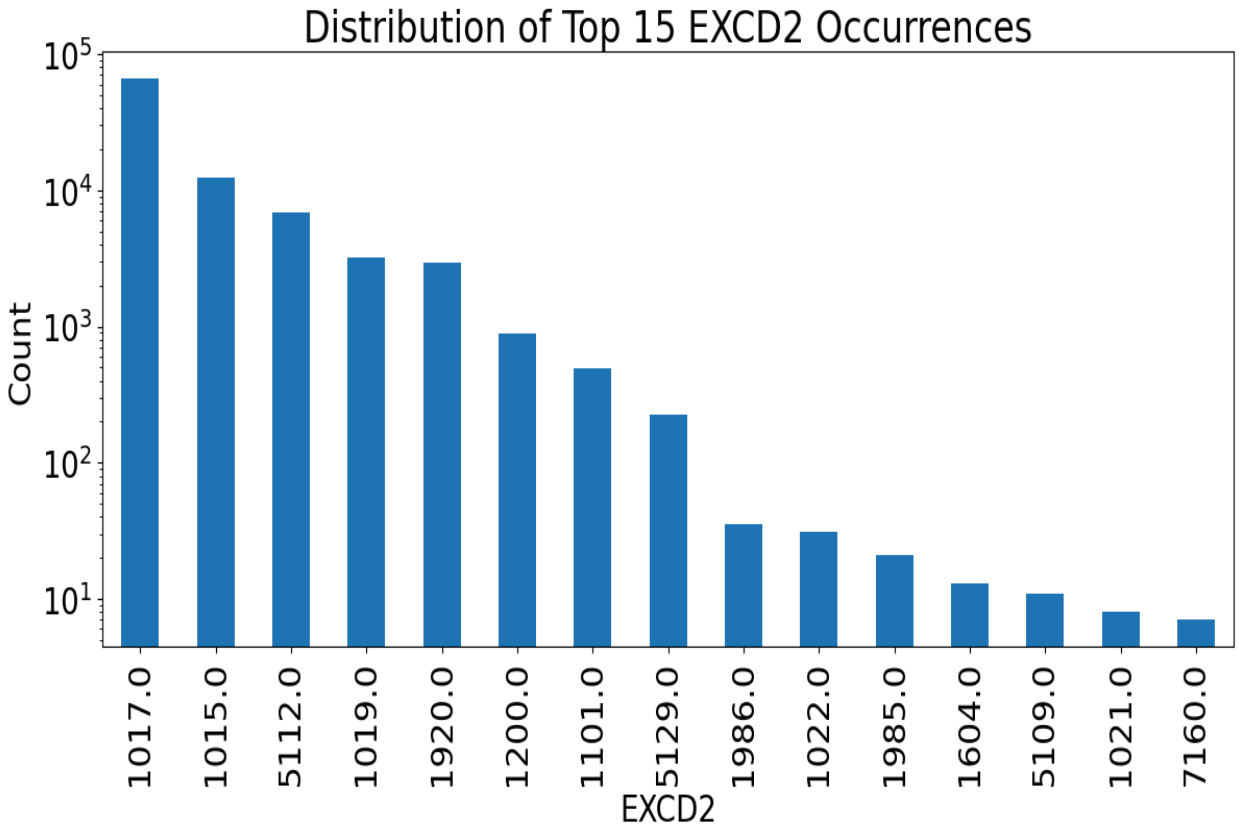
The ZIP field represents the ZIP codes of properties. The chart highlights the top 20 most frequent ZIP codes, with 10314.0 having the highest count (26,066 records), followed by 11234.0 (20,021) and 10301.0 (18,927). The remaining ZIP codes show relatively even distributions, indicating a broad geographical spread of properties across the dataset.

(14) Field Name: EXMPTCL



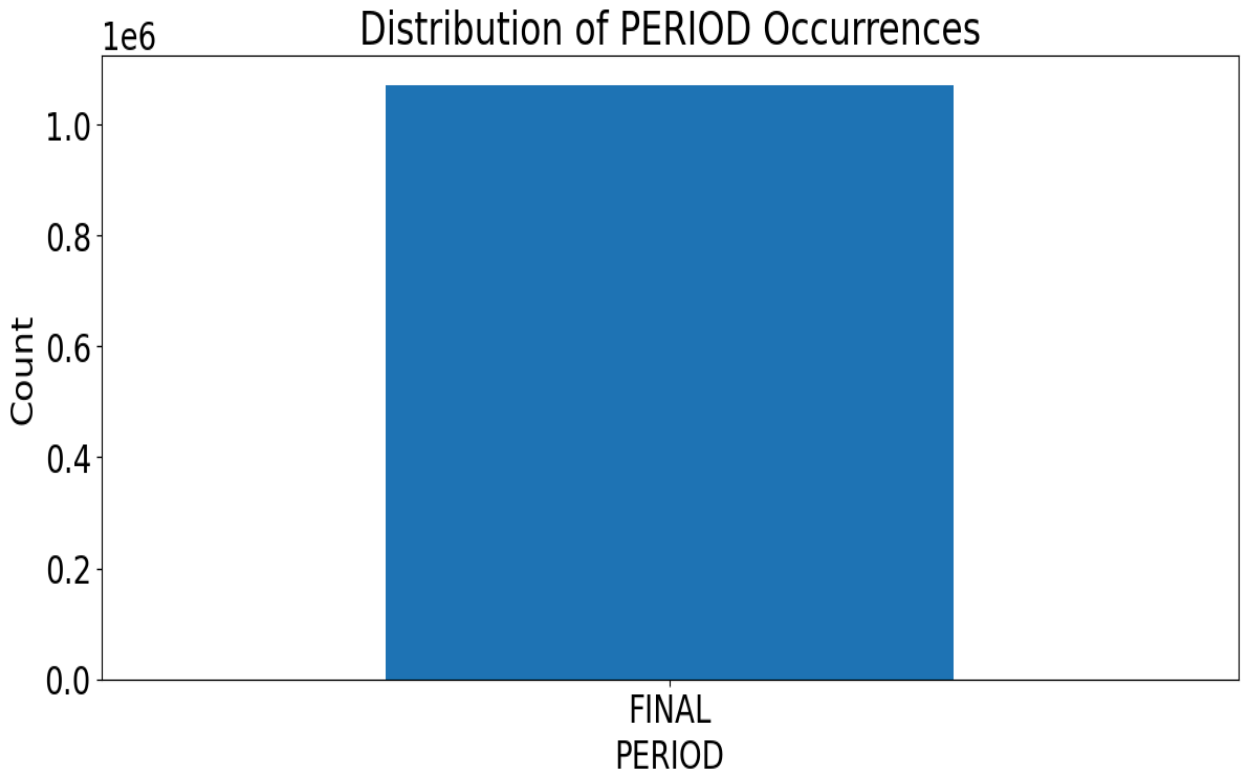
The EXMPTCL field represents exemption classifications for properties. The chart shows the top 14 exemption classes, with X1 being the most frequent (6,912 records), followed by X5 (5,208) and X7 (820). Other classes, such as X8 and X3, have fewer records, while the remaining categories, including VI and KI, are relatively rare. This field provides insights into property tax exemptions and classifications.

(15) Field Name: EXCD2



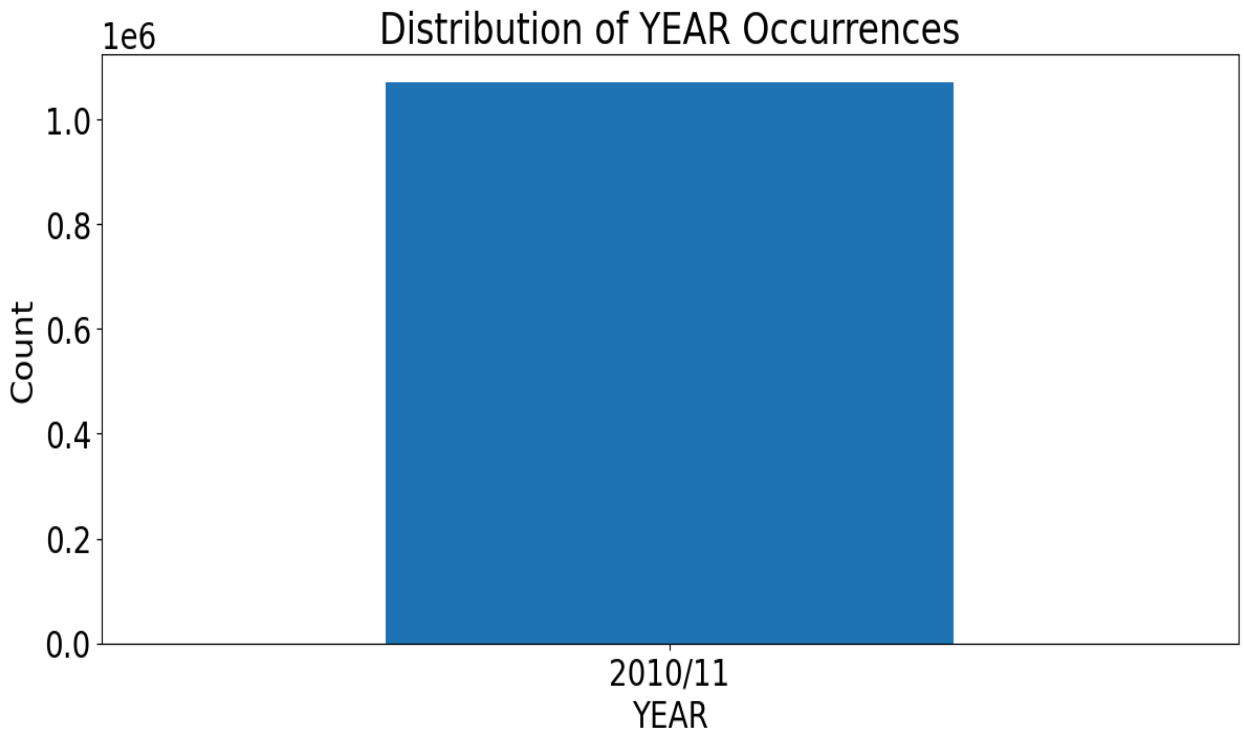
The EXCD2 field represents secondary exception codes for properties. The chart highlights the top 15 most frequent codes, with 1017.0 being the most common, followed by 1015.0 and 5112.0. The remaining codes show a gradual decline in frequency, reflecting the diversity of exception classifications within the dataset.

(16) Field Name: PERIOD



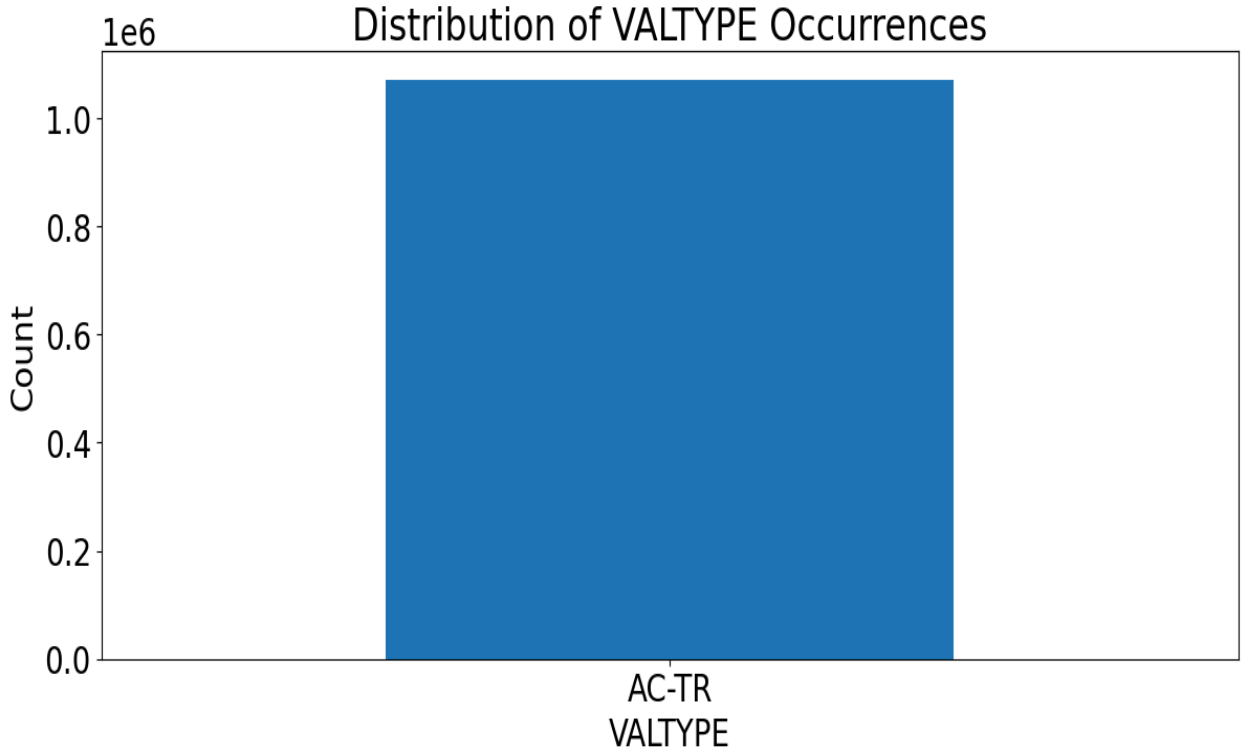
The PERIOD field represents the time period for the dataset. The chart shows that all records fall under the value FINAL, indicating uniformity across the dataset for this field. This consistency suggests that the data pertains to a specific finalized period, making it less relevant for variability analysis.

(17) Field Name: YEAR



The YEAR field represents the dataset's reference year. All records fall under the value 2010/11, indicating that the data pertains exclusively to this specific year. This uniformity suggests the dataset is limited to a single time frame, making this field non-variable.

(18) Field Name: VALTYPE



The VALTYPE field represents the valuation type for properties. All records in the dataset have the value AC-TR, indicating uniformity across this field. This consistency suggests that the dataset focuses on a specific valuation type, making this field non-variable.